EWB

COURSES

EWB CLOUD LABS

STUDENT PROJECT REPORT

BUILD / TEST / EXPLAIN

AWS NETWORKING LAB

VPC Peering

Connect two VPCs.

Prove private connectivity.

EC2 workloads • custom route tables • security groups • bastion access

AWS ARCHITECTURE ASSOCIATE — STUDENT PROJECT

Name: Md Saqeeb Afridi

Batch: Batch 1

Region: Asia Pacific (Mumbai) — ap-south-1

Completion date: 23 October 2026

LEARN. ELEVATE. EMPOWER.

1. Project Summary

This project builds two separate Amazon VPC networks and connects them with a VPC peering connection. Private EC2 workloads communicate using private IPv4 addresses; routing and security-group rules control the permitted traffic. A bastion host in VPC A provides the administrative path to the private workload in VPC B.

Field	Configured value
Student	Md Saqeeb Afridi
Batch	Batch 1
AWS Region	Asia Pacific (Mumbai), ap-south-1
VPC A	ewb-vpc-a-lab — 10.30.0.0/16
VPC B	ewb-vpc-b-lab — 10.40.0.0/16
Peering connection	ewb-a-b-peer — pcx-0664d194a986e4999 (Active)
App-A private IPv4	10.30.2.58
App-B private IPv4	10.40.1.117
Bastion public / private IPv4	13.201.10.81 / 10.30.1.177

Evidence note: Values above come from the AWS console details and terminal results visible in the conversation. Items not evidenced in the available captures are marked as pending verification rather than claimed complete.

2. Architecture and Resource Inventory

The two VPC CIDR blocks do not overlap, allowing the peering connection to route between them. Peering is not a transit gateway: it does not automatically provide transitive routing through another VPC.

Resource	Name / ID	Address / purpose
VPC A	ewb-vpc-a-lab	10.30.0.0/16
VPC B	ewb-vpc-b-lab	10.40.0.0/16
Subnet A public	ewb-a-public	10.30.1.0/24
Subnet A private	ewb-a-private	10.30.2.0/24
Subnet B private	ewb-b-private	10.40.1.0/24
Route table A public	rtb-0a3774ef3e1d6415f	Default internet route plus peering route to 10.40.0.0/16
Route table A private	rtb-0f9a059f2ad95d2dc	10.40.0.0/16 → peering
Route table B private	rtb-08a3e6249bd87f1cf	10.30.0.0/16 → peering
Peering	pcx-0664d194a986e4999	Active

Subnet associations should be verified in the AWS console and included in the final evidence set. The public subnet belongs to VPC A; both application workloads use private subnets and private IPv4 destinations.

3. Route Tables and Traffic Path

Required peering routes

Route table	Destination	Target
ewb-rt-a-private	10.40.0.0/16	pcx-0664d194a986e4999
ewb-rt-b-private	10.30.0.0/16	pcx-0664d194a986e4999
ewb-rt-a-public	10.40.0.0/16	pcx-0664d194a986e4999 (for bastion-originated traffic)

Forward path: App-B (10.40.1.117) sends traffic to App-A (10.30.2.58). App-B’s subnet route table sends 10.30.0.0/16 to the peering connection. The peering connection delivers the packet to VPC A, where the destination subnet route table handles the local destination.

Return path: App-A’s subnet route table sends traffic for 10.40.0.0/16 through the same peering connection. Both directions need valid routes and security-group permission; one-sided routing is insufficient.

Bastion administration: the Windows SSH connection used the bastion at 13.201.10.81 as a jump host to reach App-B privately. Do not include private keys, passwords, or access tokens in the submitted report.

4. Security Groups and Connectivity Tests

Security groups should allow only the traffic needed for the lab. The captured App-A rules included ICMP from 10.40.1.0/24 and from the bastion private IP 10.30.1.177/32, plus SSH access from the bastion security group. Verify the exact inbound and outbound rules for all three instances before submission.

Connectivity evidence observed

Test	Observed result	Meaning
Bastion → App-B private IP 10.40.1.117 (ping)	4/4 replies in the captured terminal	ICMP path worked at test time
Bastion → App-B TCP/22	TCP port 22 reachable	SSH network path available
Windows SSH via ProxyCommand → App-B	Shell prompt ec2-user@ip-10-40-1-117 shown	Private SSH access succeeded
App-B → App-A 10.30.2.58 (ping)	4 packets received, 0% loss on successful retry	Peering and return path worked
Earlier App-B → App-A attempt	100% packet loss	Initial failure; later successful retry confirms recovery, but controlled rule-change cause needs explicit evidence

Important: the current evidence confirms successful connectivity, but a controlled ICMP failure-and-recovery test should show the deliberate security-group change, the failed ping, the restored rule, and the successful ping. Do not claim the failure was deliberately induced unless those steps were actually performed and captured.

5. IAM Access Analyzer and Credential Report

An IAM Access Analyzer was created in the Mumbai region with the finding type “Resource analysis – External access”. The console displayed a success banner confirming creation. The findings page initially showed zero findings; AWS notes that findings may take time to become available. Zero findings at that moment does not prove that no external access exists.

The IAM Credential Report was generated and downloaded from the global IAM console. Keep the downloaded CSV/XLSX report as a separate supporting artifact if your instructor requests it. Do not expose access keys or sensitive account information in screenshots or in the final report.

Access Analyzer screenshot and credential-report screenshot should be included as supporting IAM evidence only if these tasks are part of the assigned workbook. The main submission checklist shown in the workbook emphasizes VPC architecture, route tables, security groups, ping/SSH tests, controlled ICMP failure/recovery, and cleanup evidence.

6. Submission Checklist and Remaining Evidence

This report organizes the information currently available. Before submitting, check each item against your own console and add the missing evidence below.

Required item	Status in this draft
Cover with name, batch, title, region, completion date	Included
Architecture diagram and inventory with resource IDs and private IPs	Inventory included; actual IDs/IPs listed where available
Subnet CIDRs and all three route tables, including associations	CIDRs/routes documented; add clear route-table and subnet-association screenshots
Active peering connection and inbound rules for each security group	Peering ID/status documented; add/verify screenshots for every SG
Baseline failure, forward/reverse ping success, and App-B SSH proof	Success and SSH details documented; include clear terminal screenshots
Controlled ICMP failure and recovery with explanation	Pending explicit evidence of deliberate rule change, failed test, restoration, and recovery
Cleanup evidence plus a short note on what was learned	Pending; capture resources deleted or describe what you intentionally retained
Credential report downloaded	Completed according to user; keep downloaded report separately if requested

Do not submit this report as fully complete until the pending evidence is added or the instructor confirms it is not required.

7. Learning Reflection

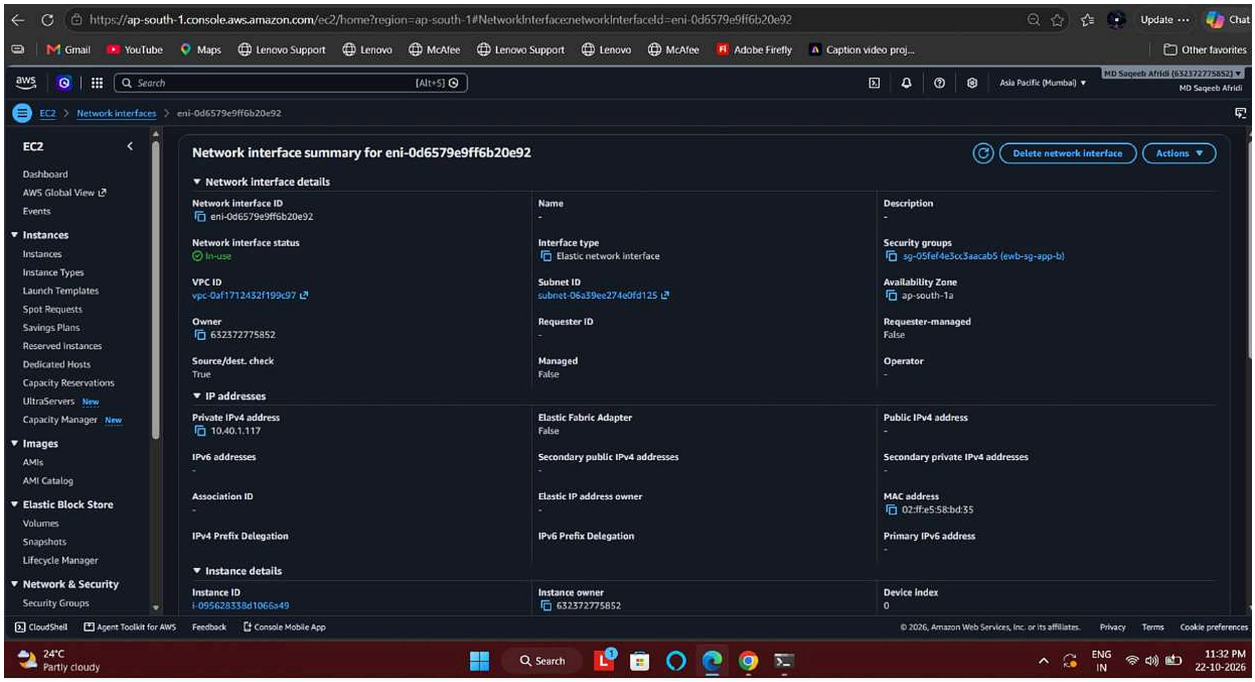
This lab demonstrates that VPC peering alone does not create end-to-end connectivity. Both VPCs need non-overlapping CIDR blocks, an accepted peering connection, routes in both directions, and security-group rules that allow the intended traffic. Private workloads can be reached securely through a bastion host without assigning public IPv4 addresses to the private application instances. Testing should include both positive and negative cases, and cleanup should be documented to avoid leaving billable resources running.

Suggested final checks

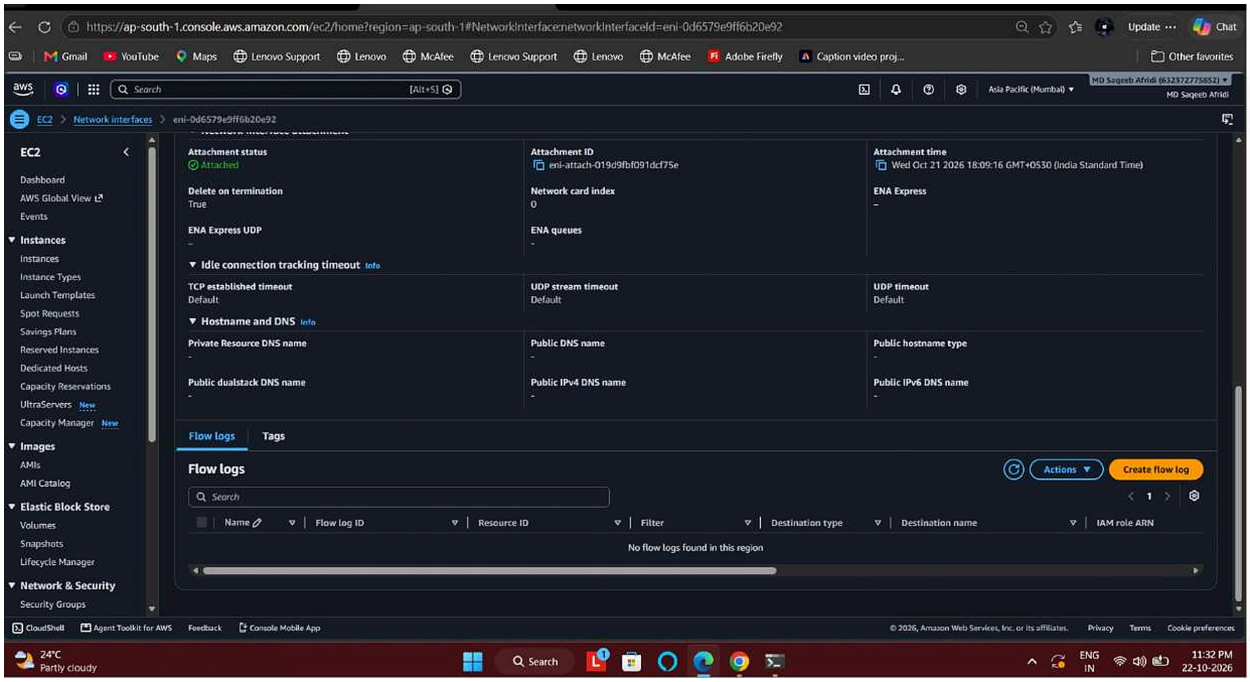
1. Verify App-A and App-B have no public IPv4 addresses. 2. Confirm both route tables point the opposite VPC CIDR at the peering connection. 3. Save readable screenshots with labels that identify host, command/test, and result. 4. Add the controlled ICMP failure/recovery evidence. 5. Capture cleanup state and state what resources were retained or deleted. 6. Export the completed report as a single PDF.

Appendix: Available screenshot captures

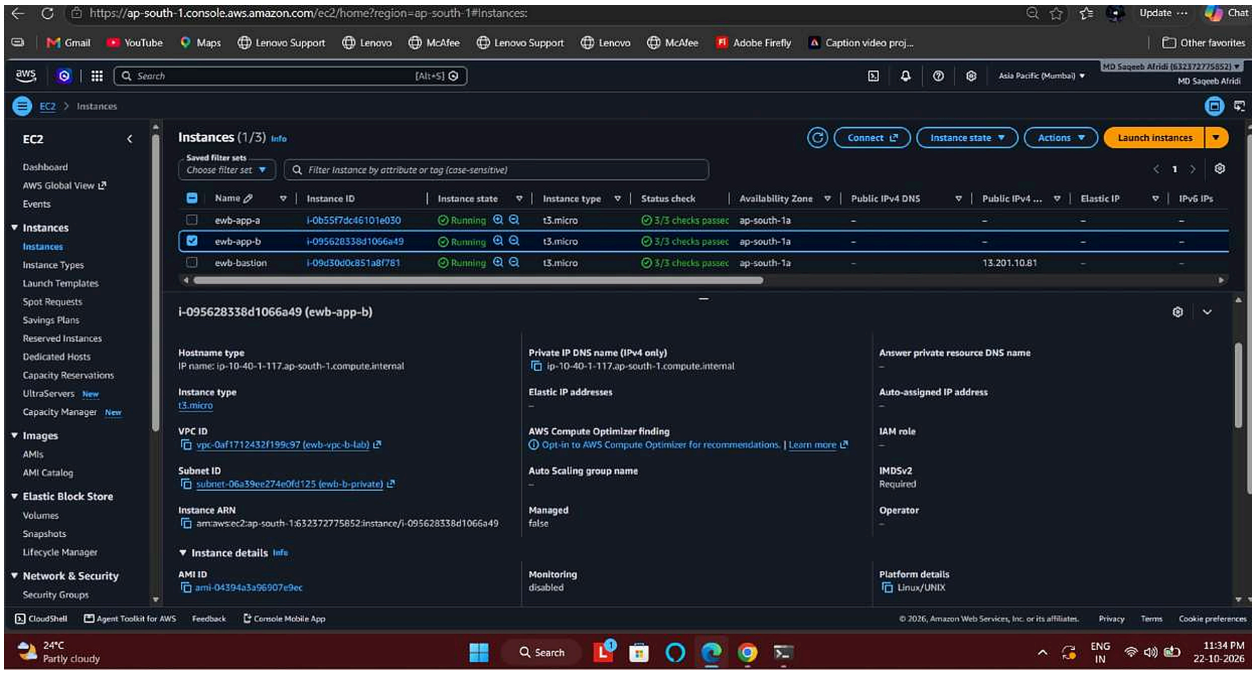
Evidence capture 01



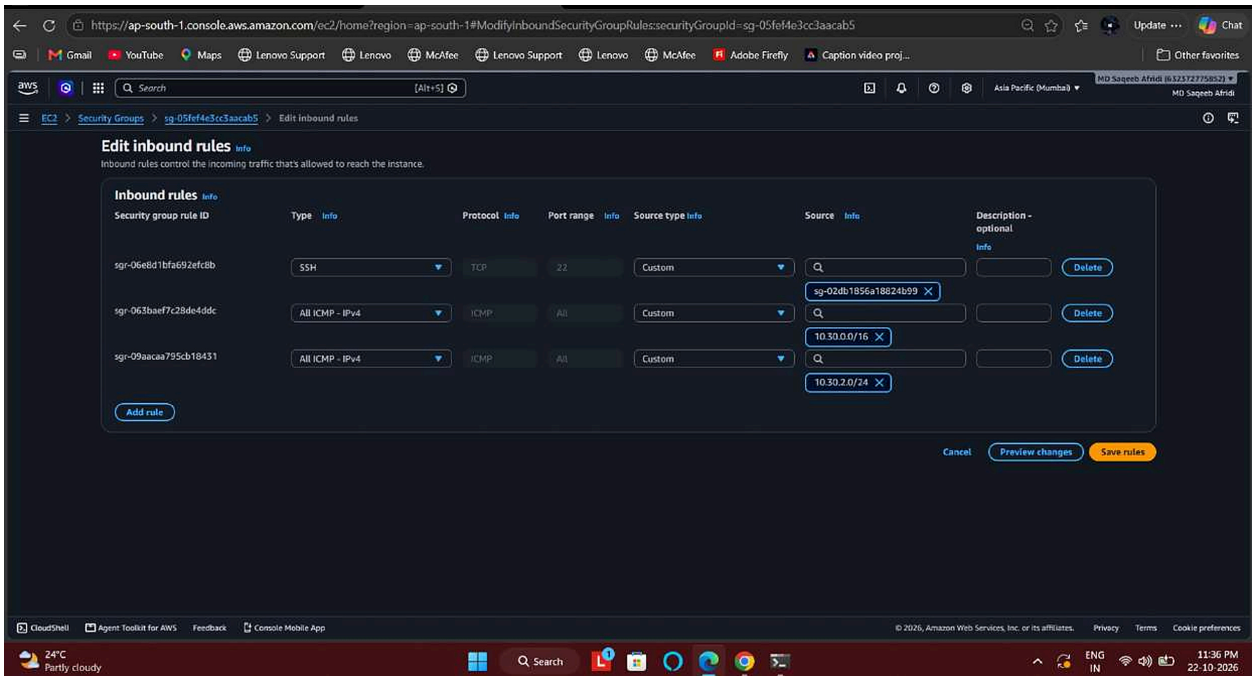
Evidence capture 02



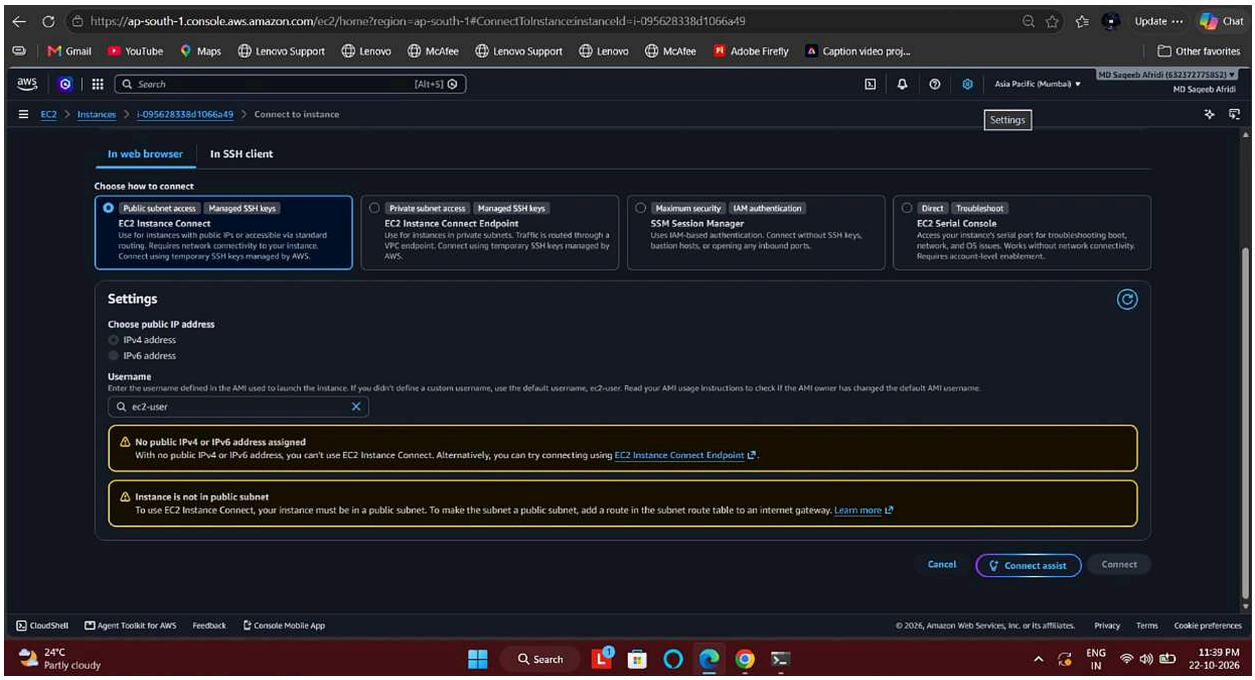
Evidence capture 03



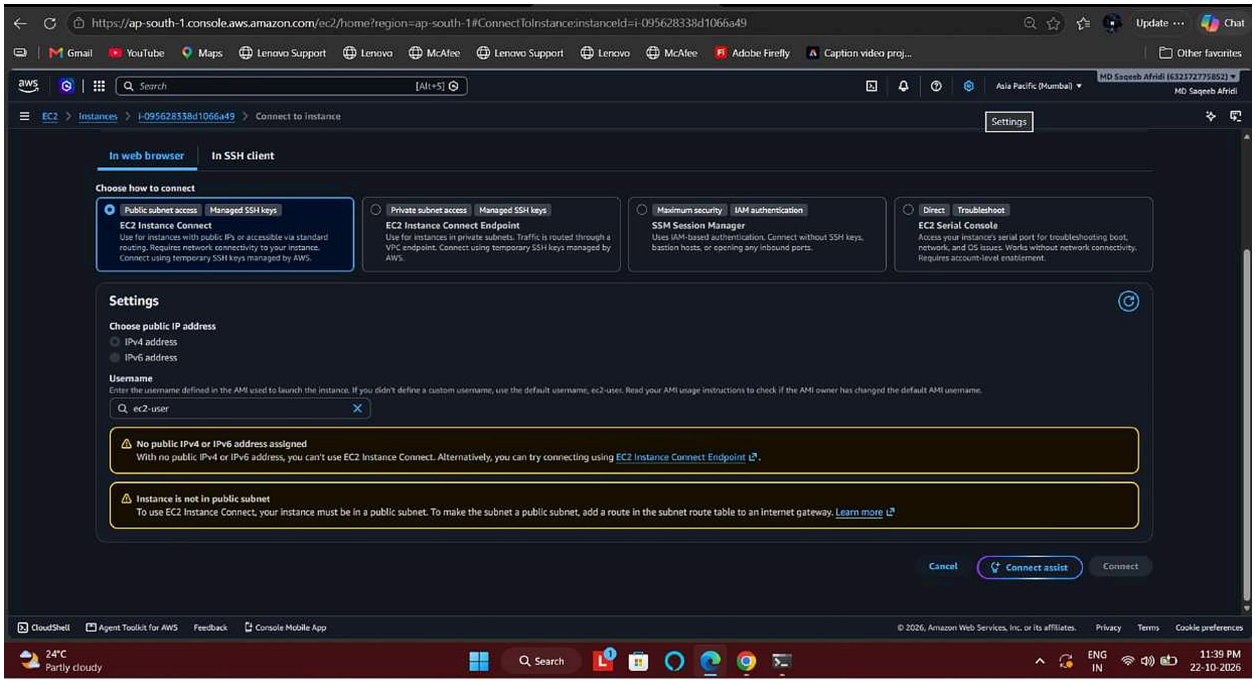
Evidence capture 04



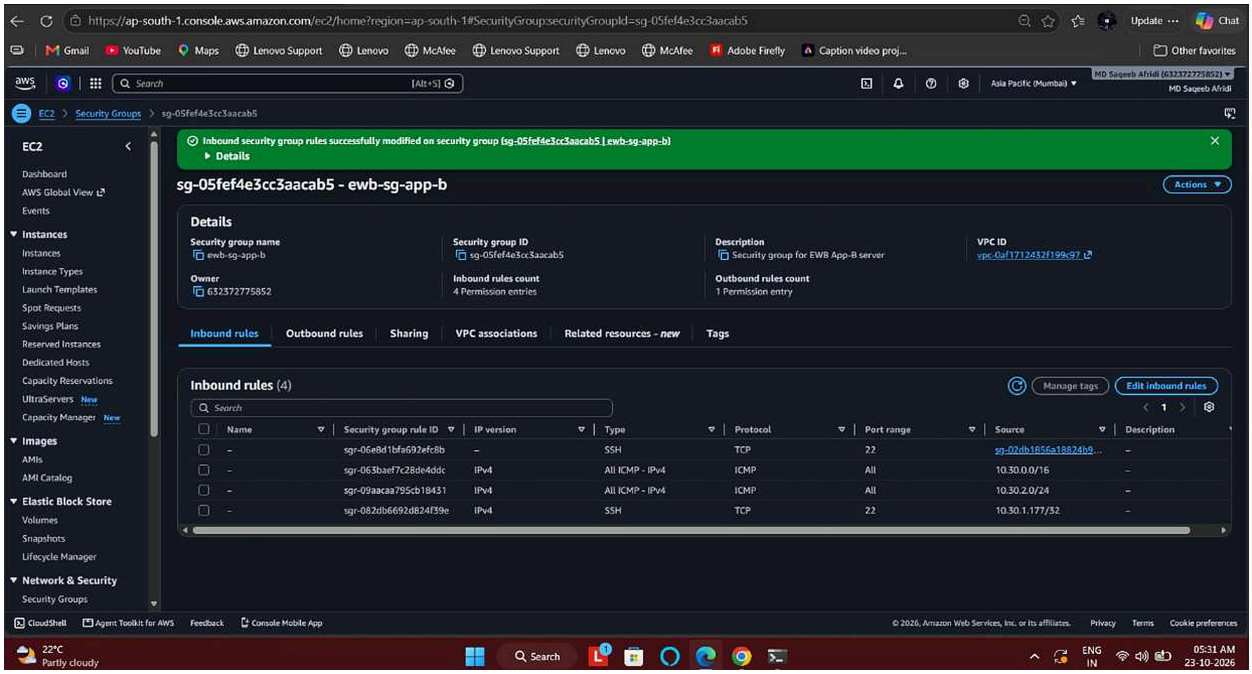
Evidence capture 05



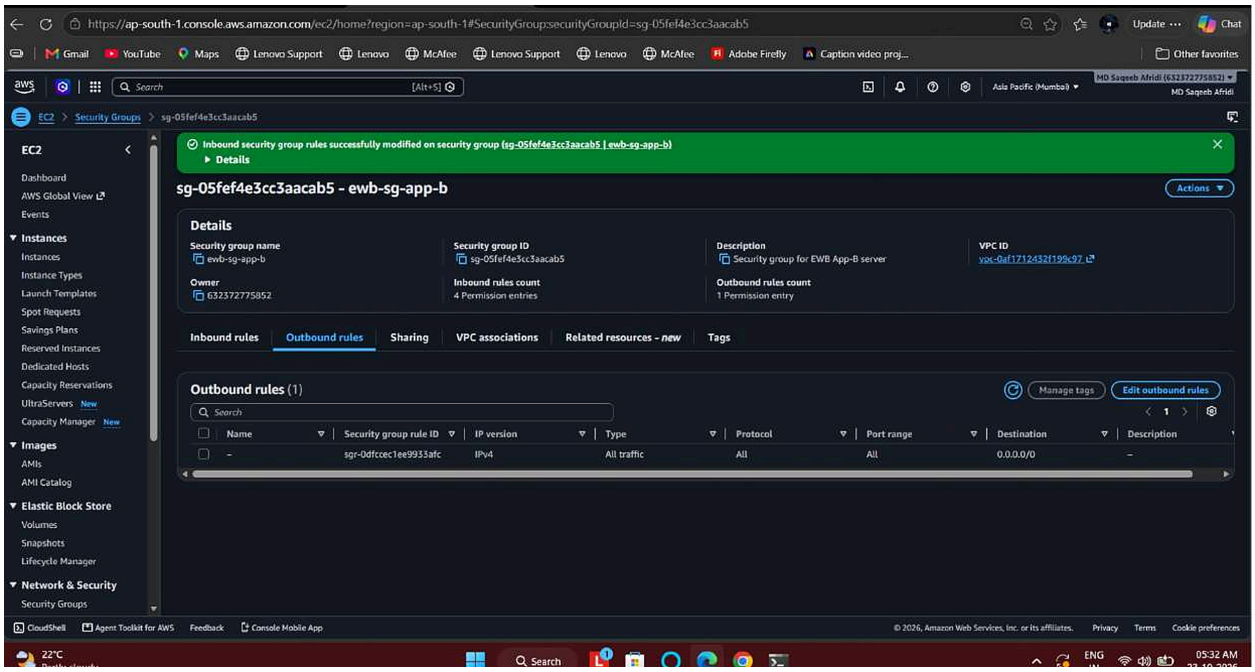
Evidence capture 06



Evidence capture 07



Evidence capture 08



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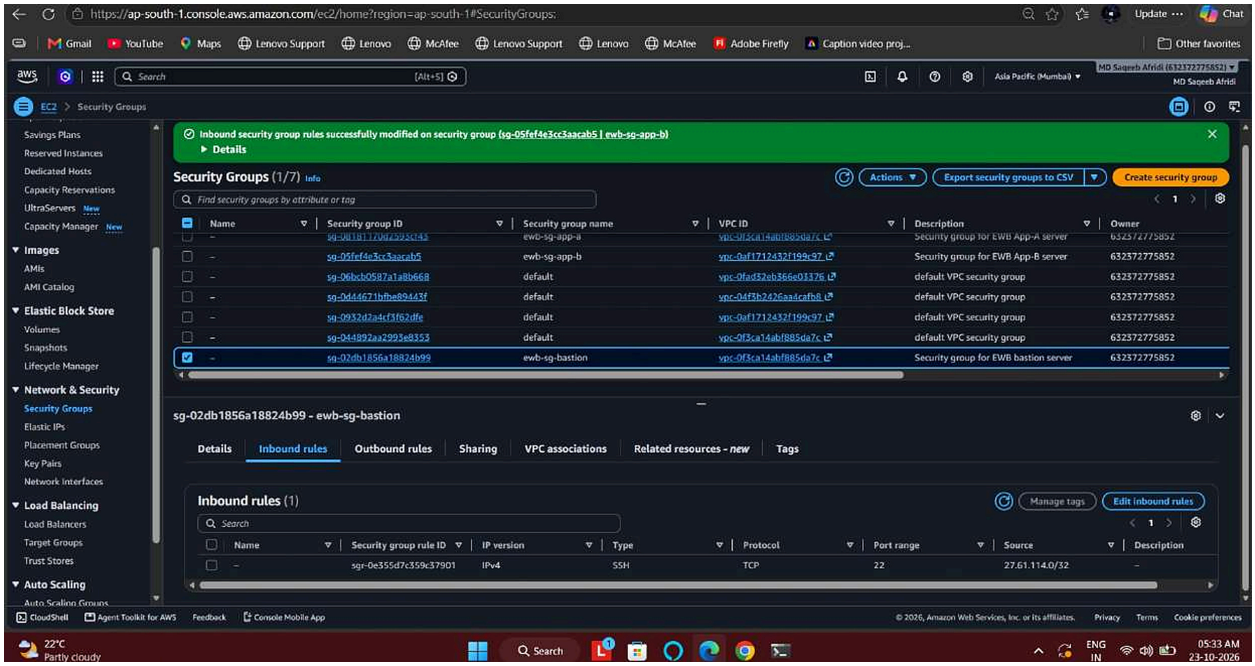
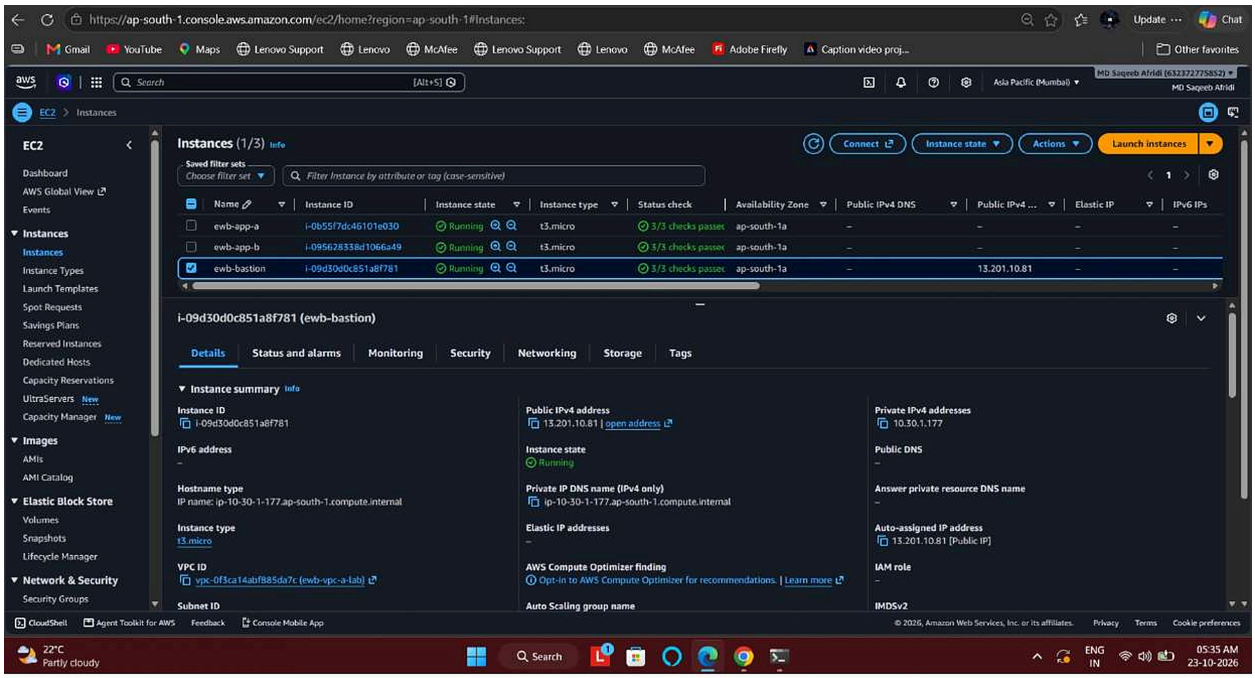
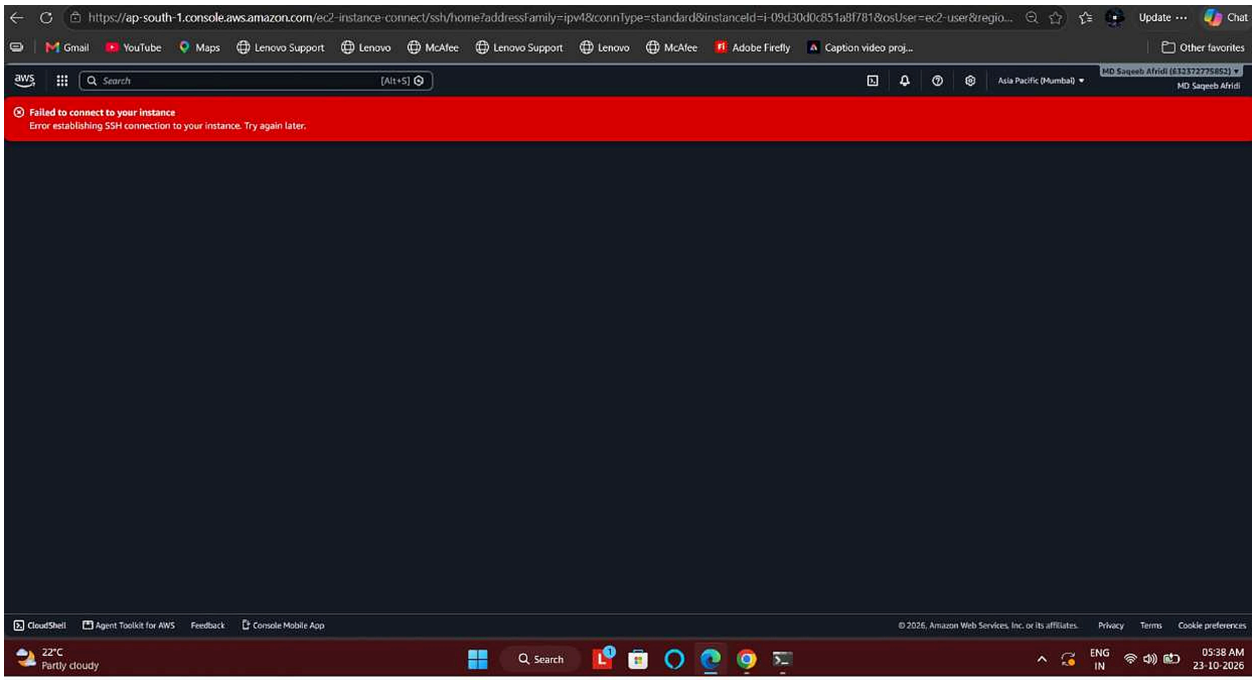


Evidence capture 10



Evidence capture 11



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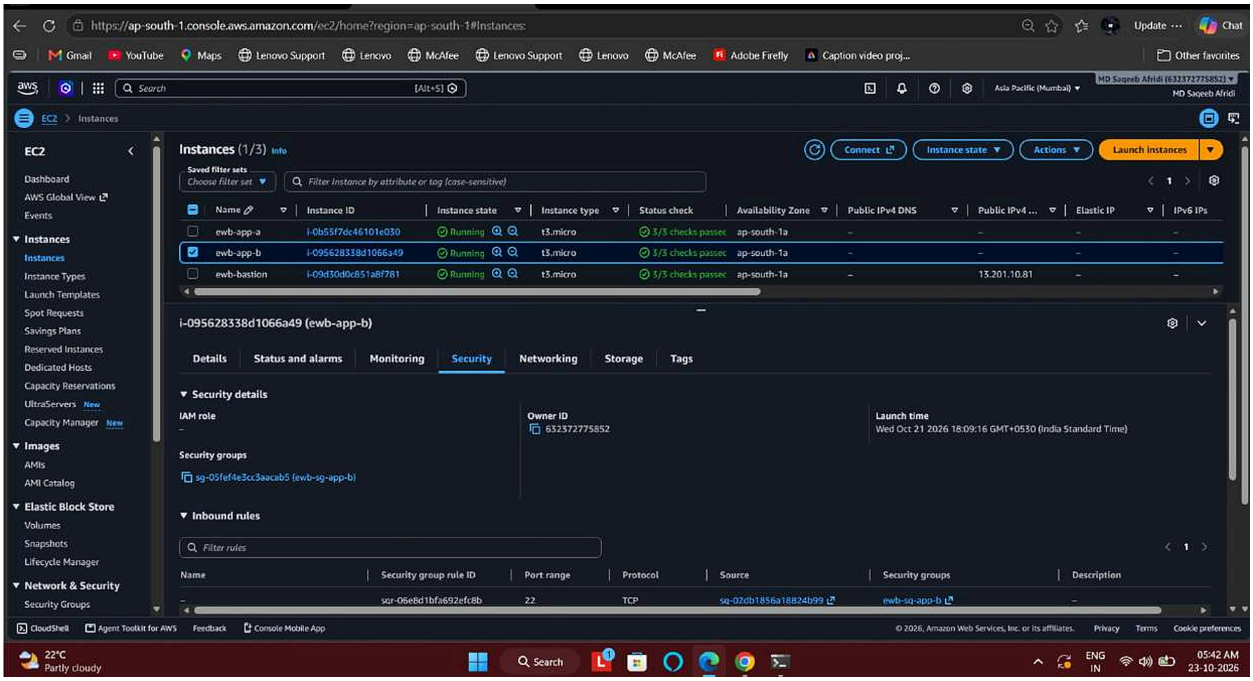
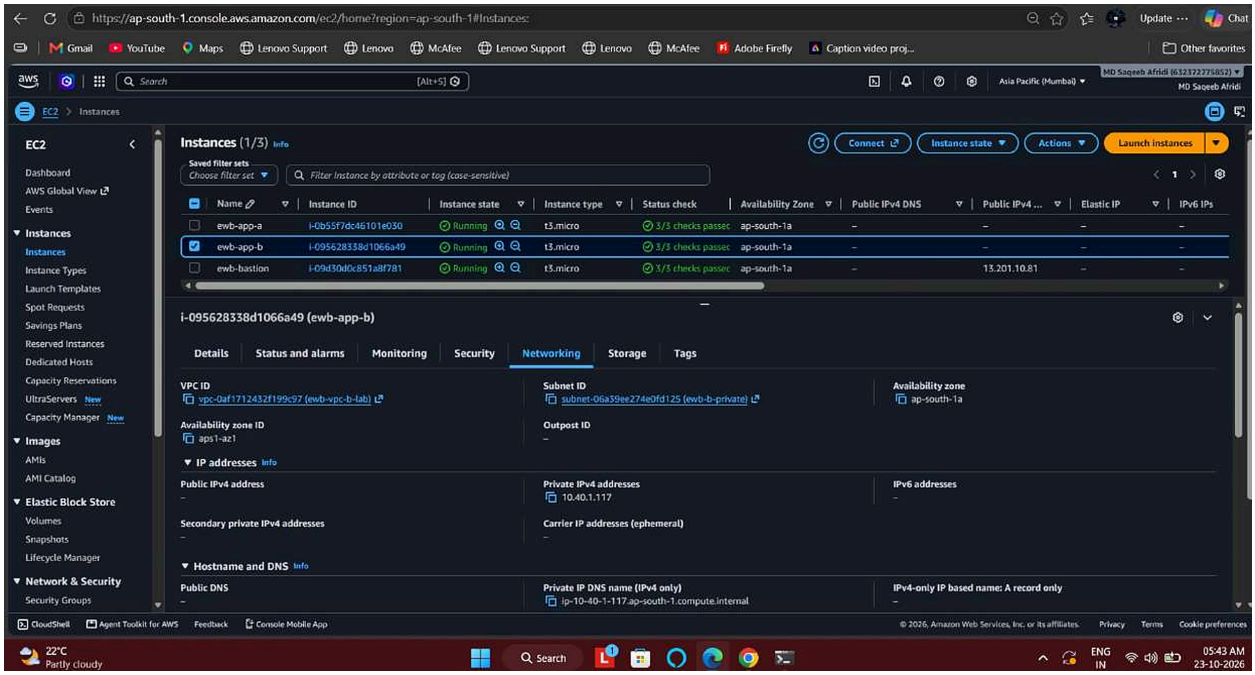


Evidence capture 13



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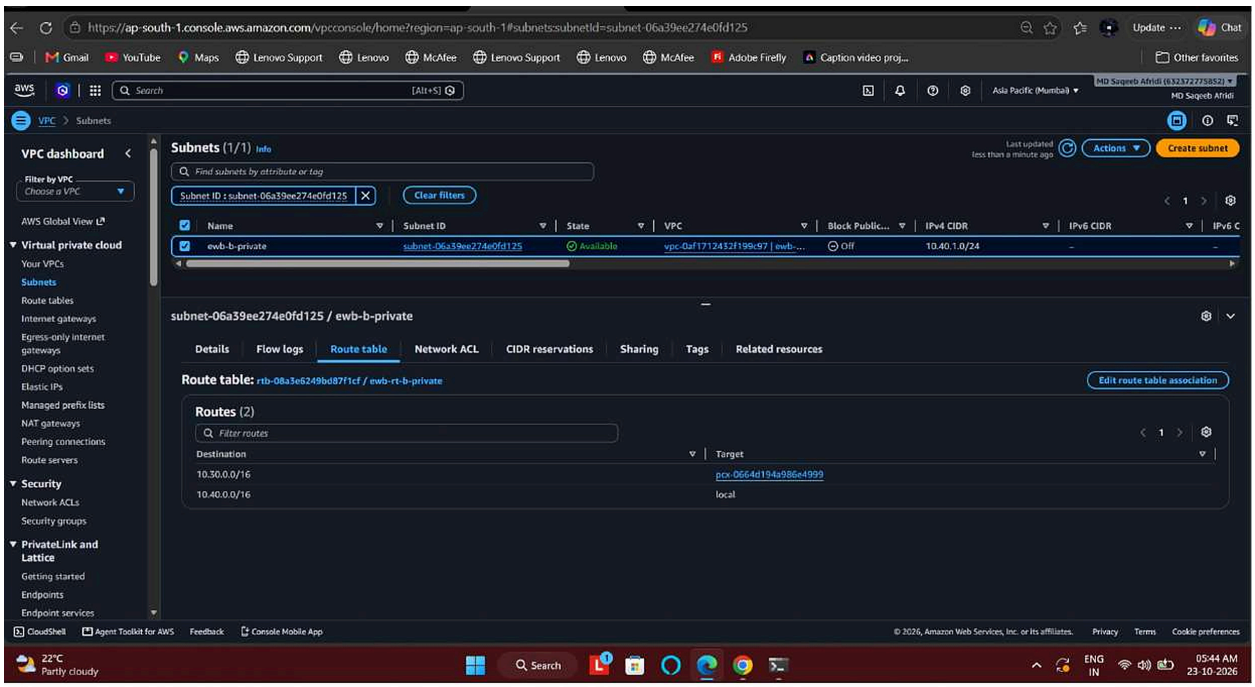

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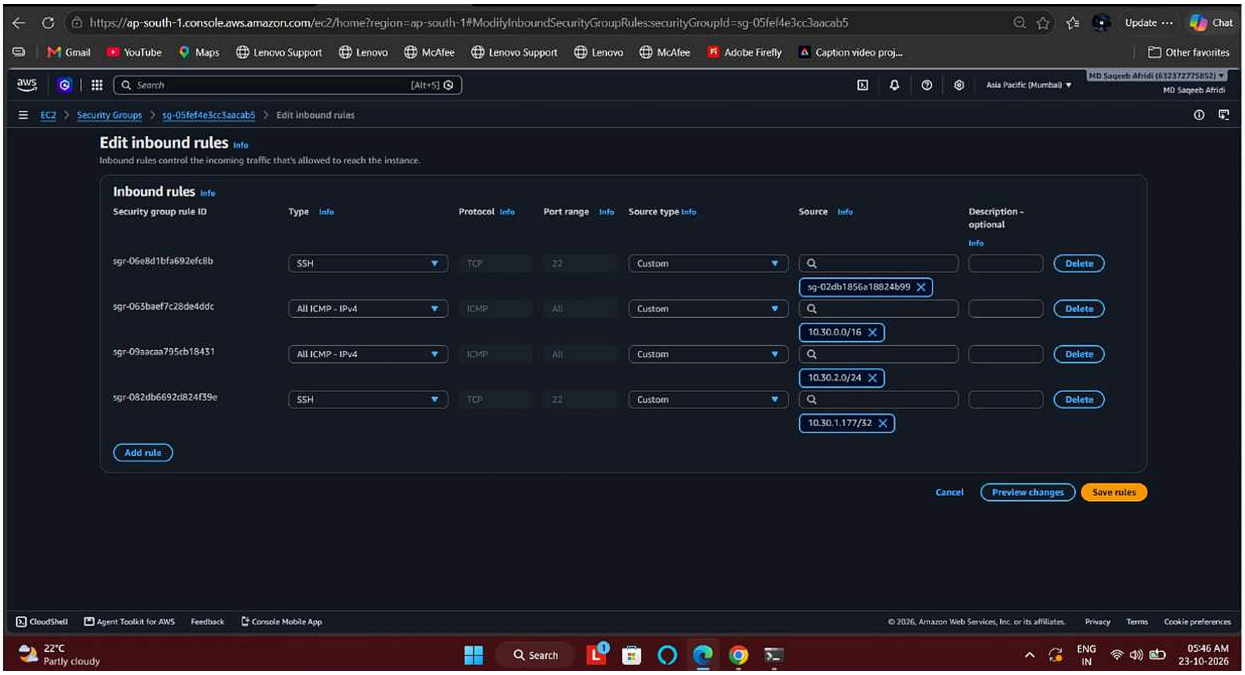

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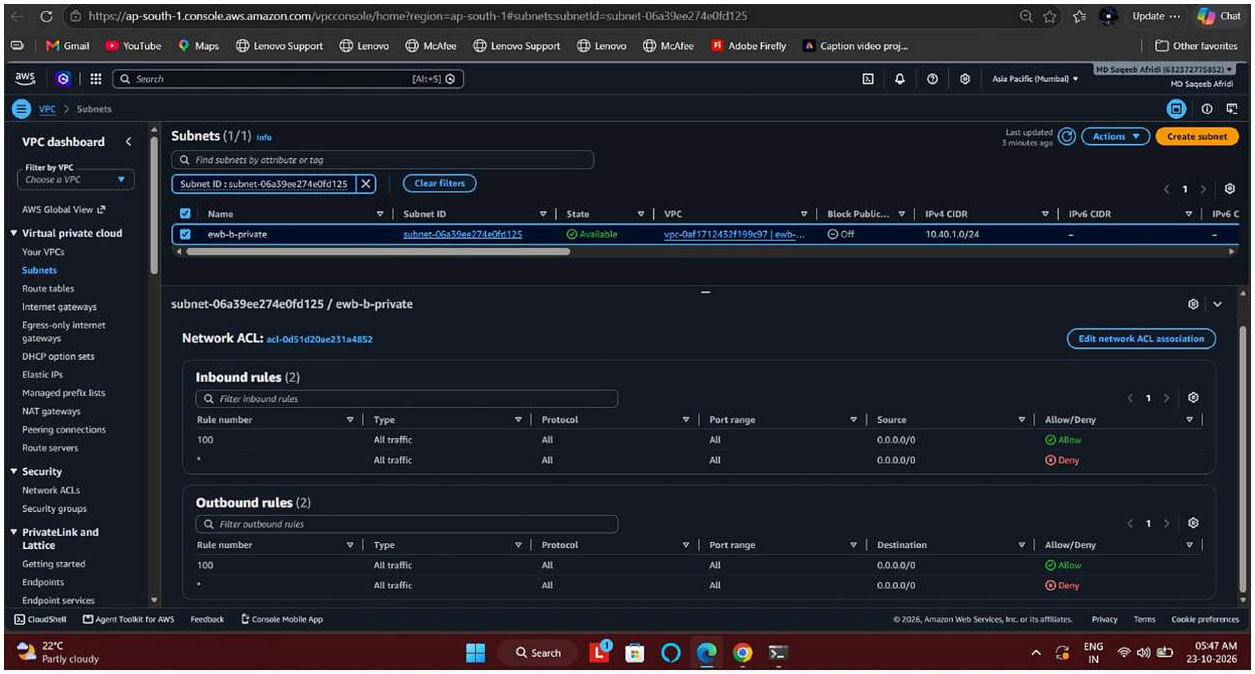

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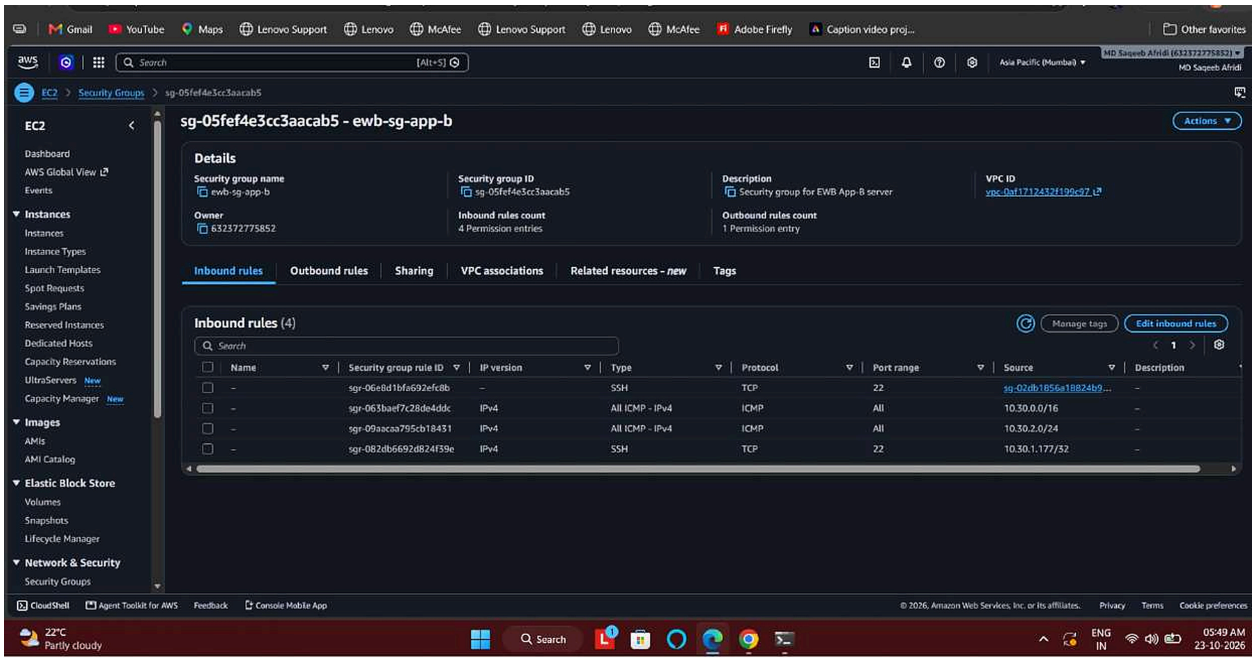
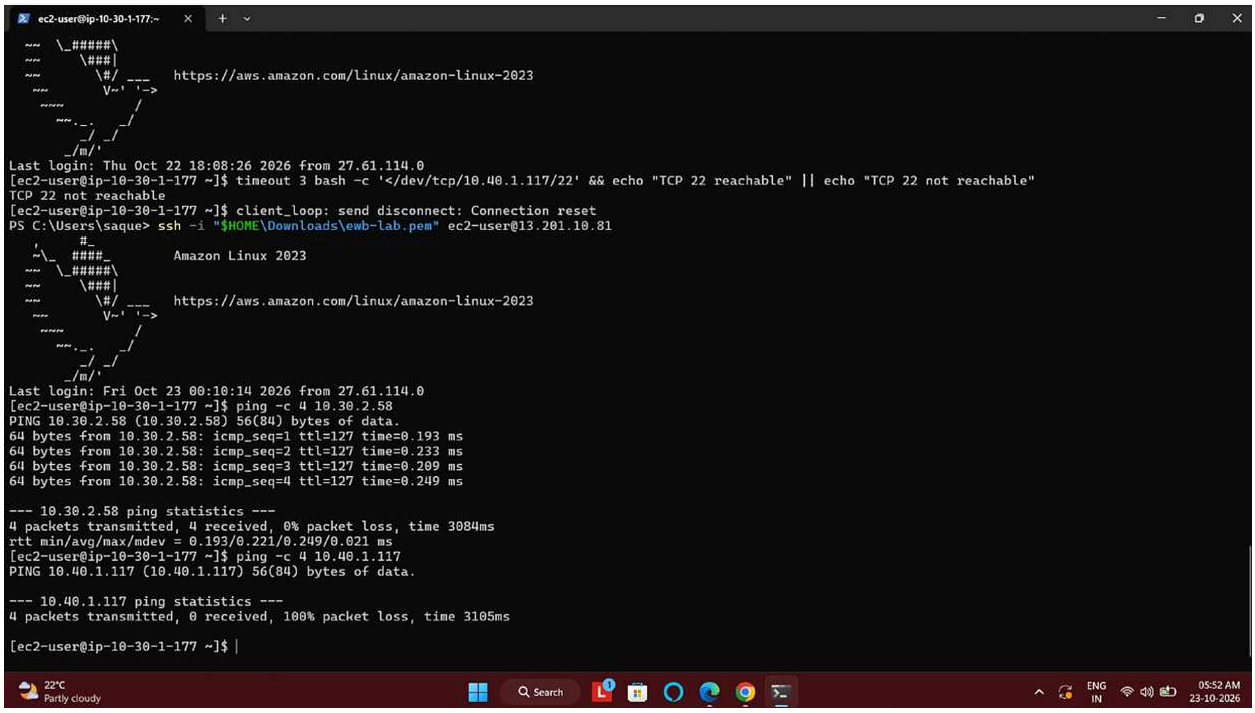


Evidence capture 18



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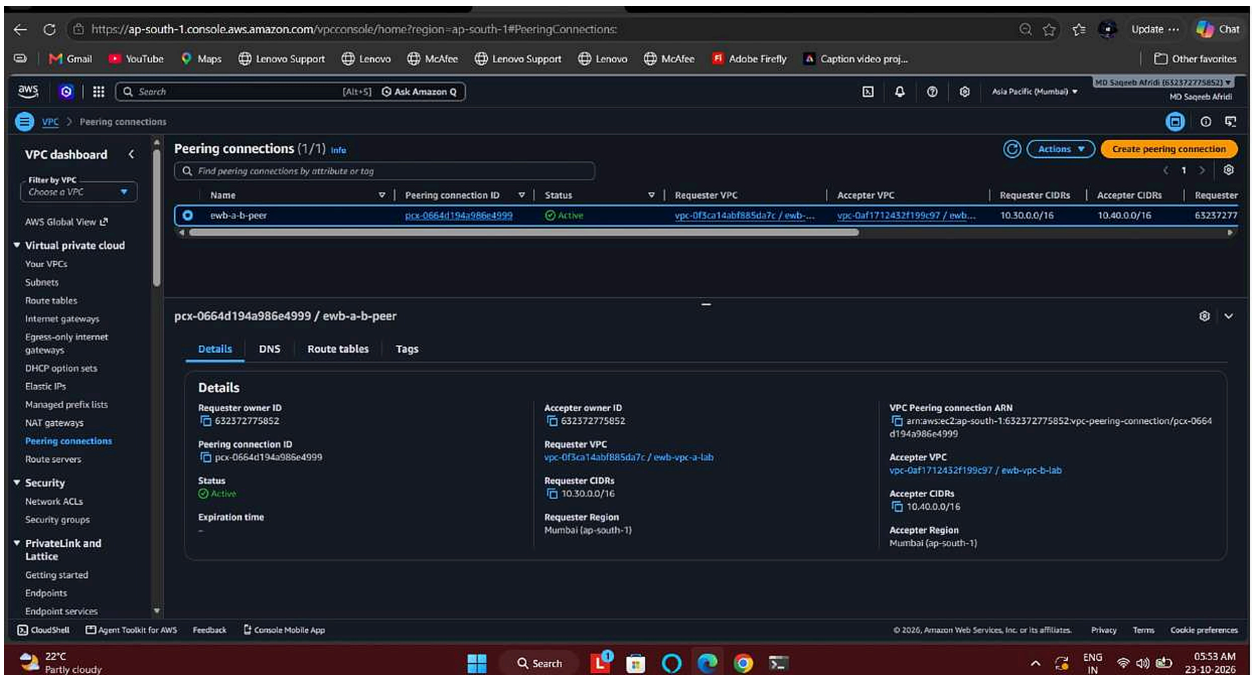

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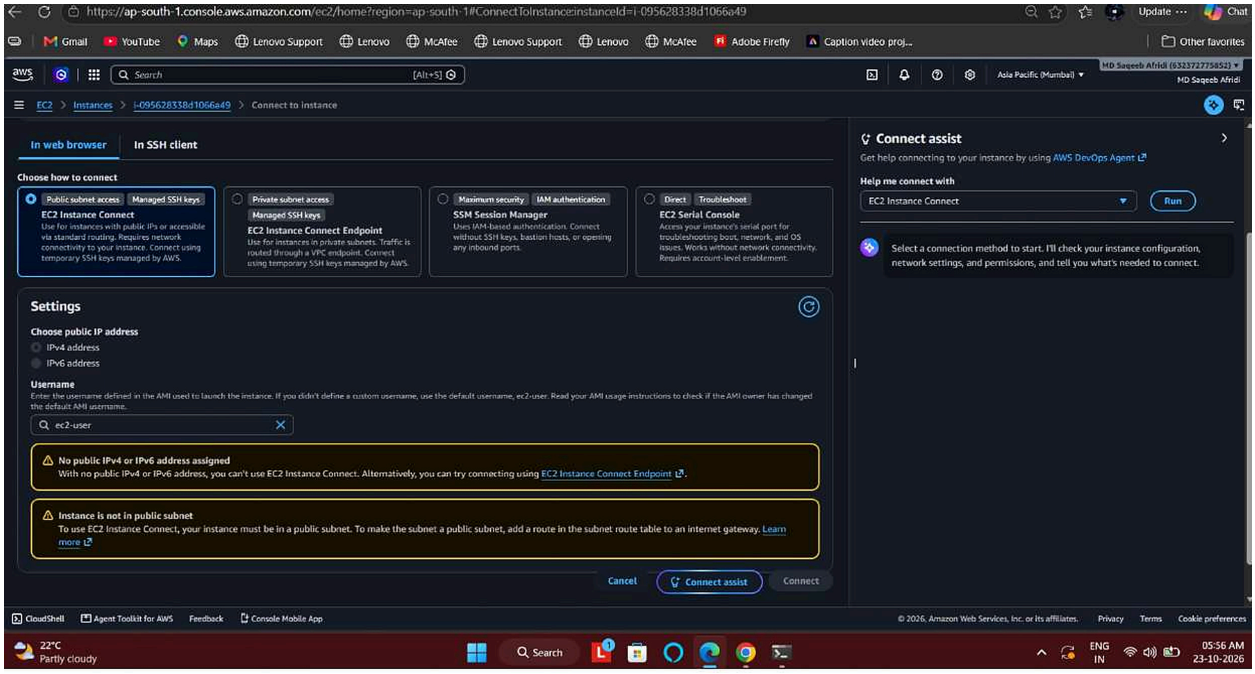

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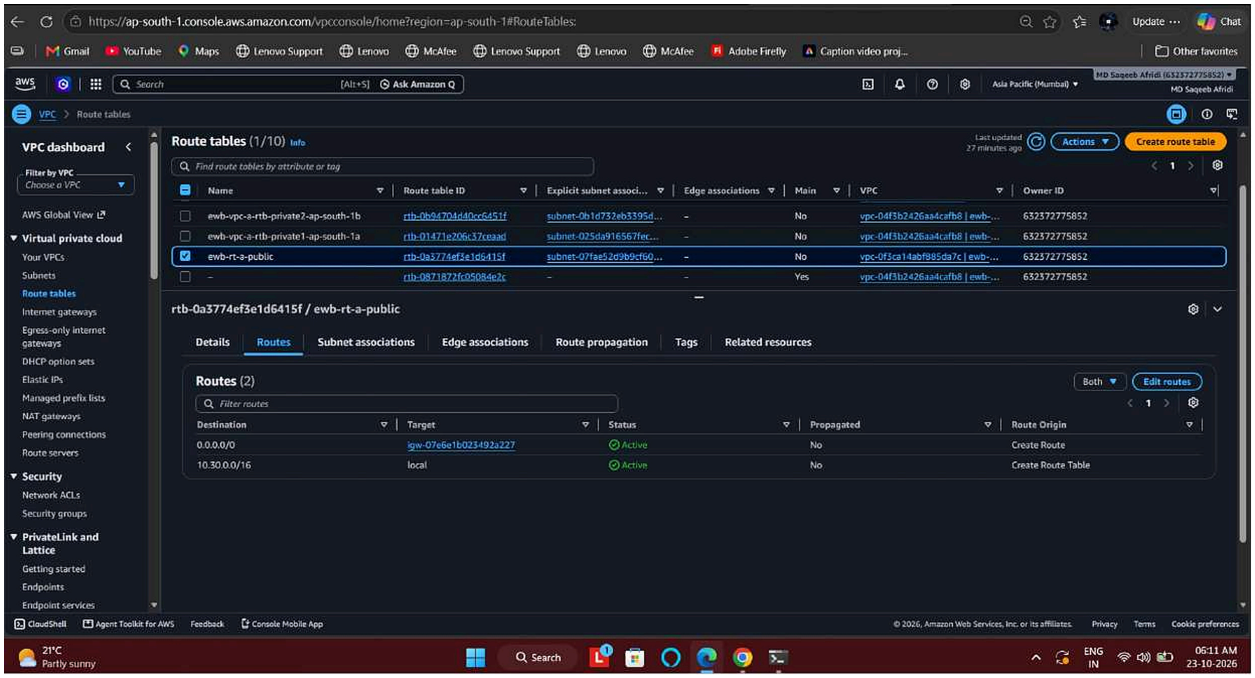

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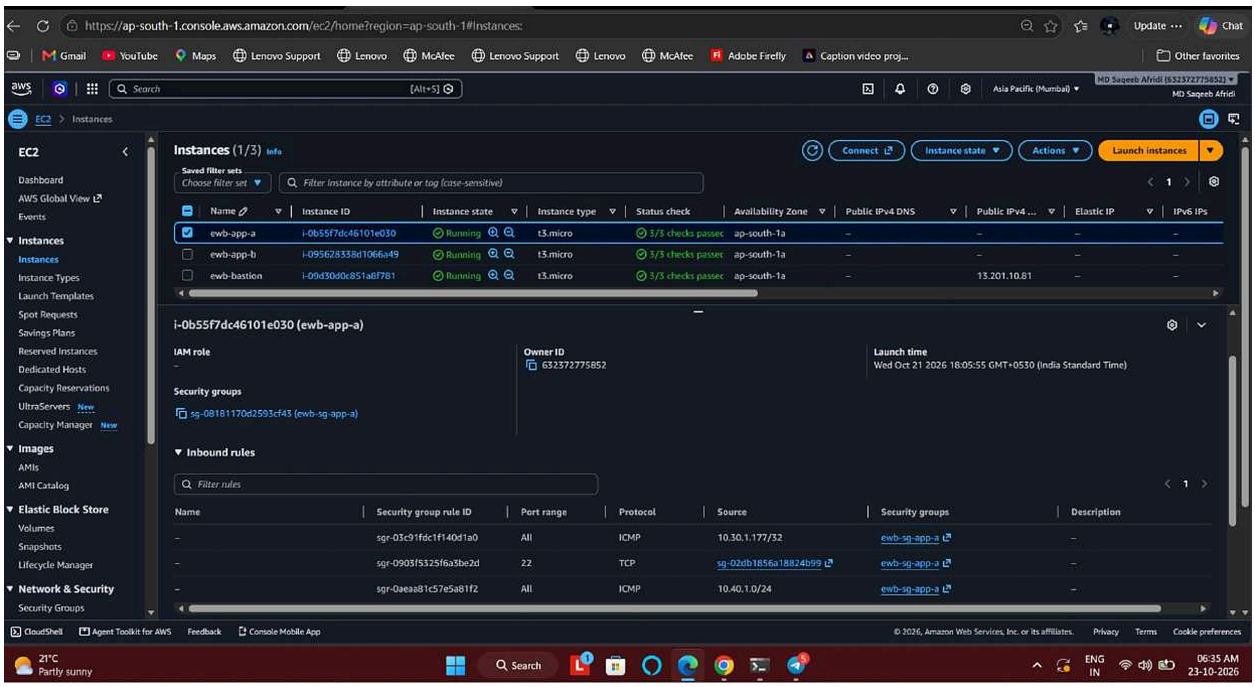

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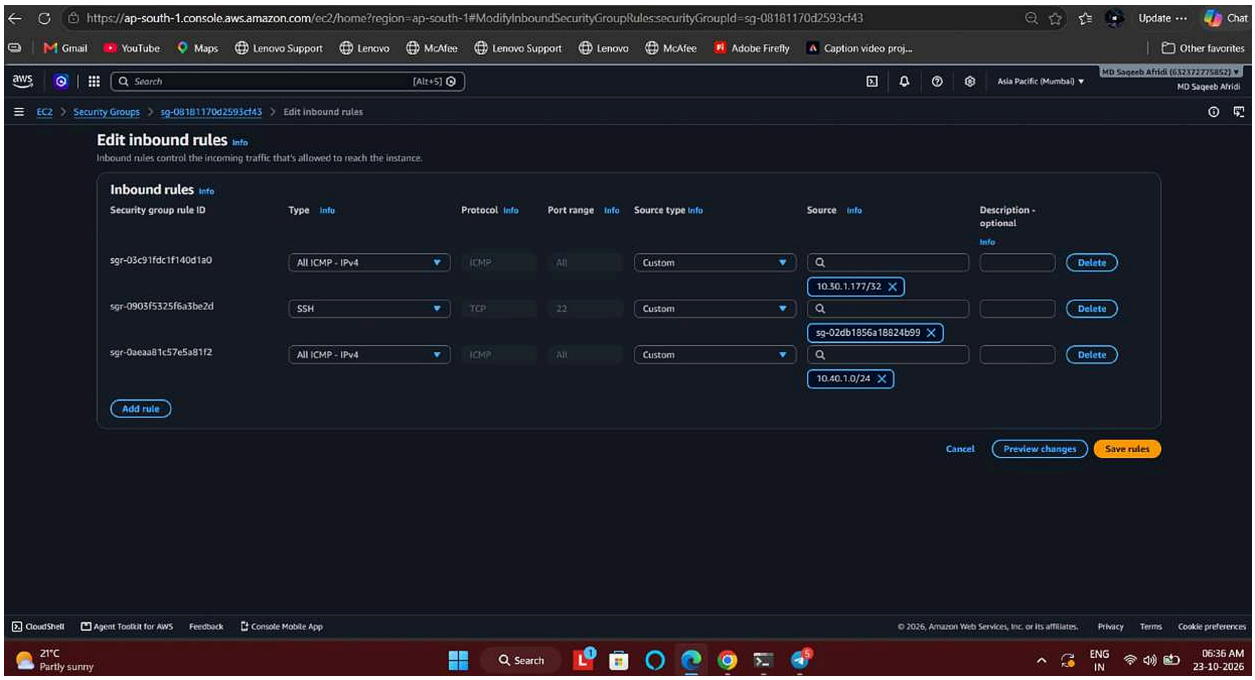

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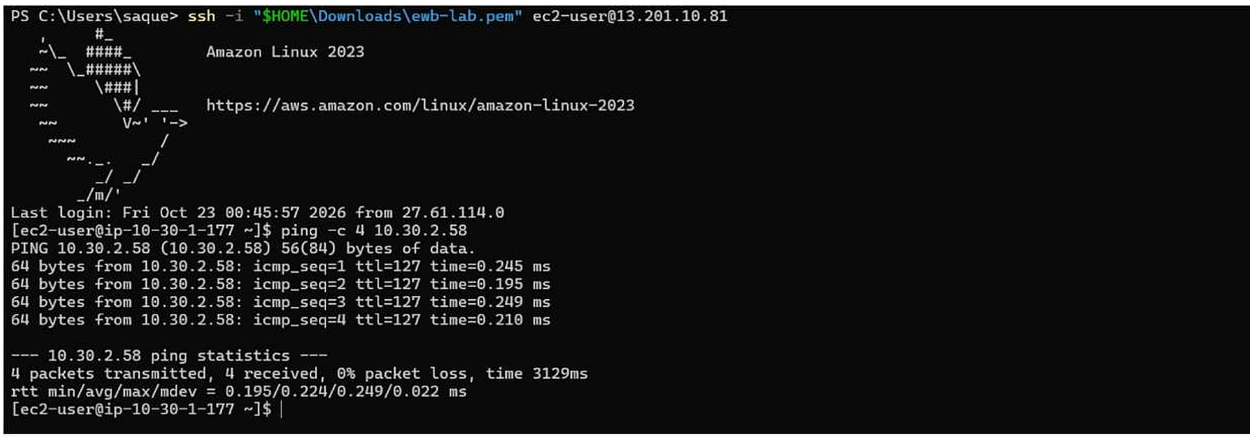

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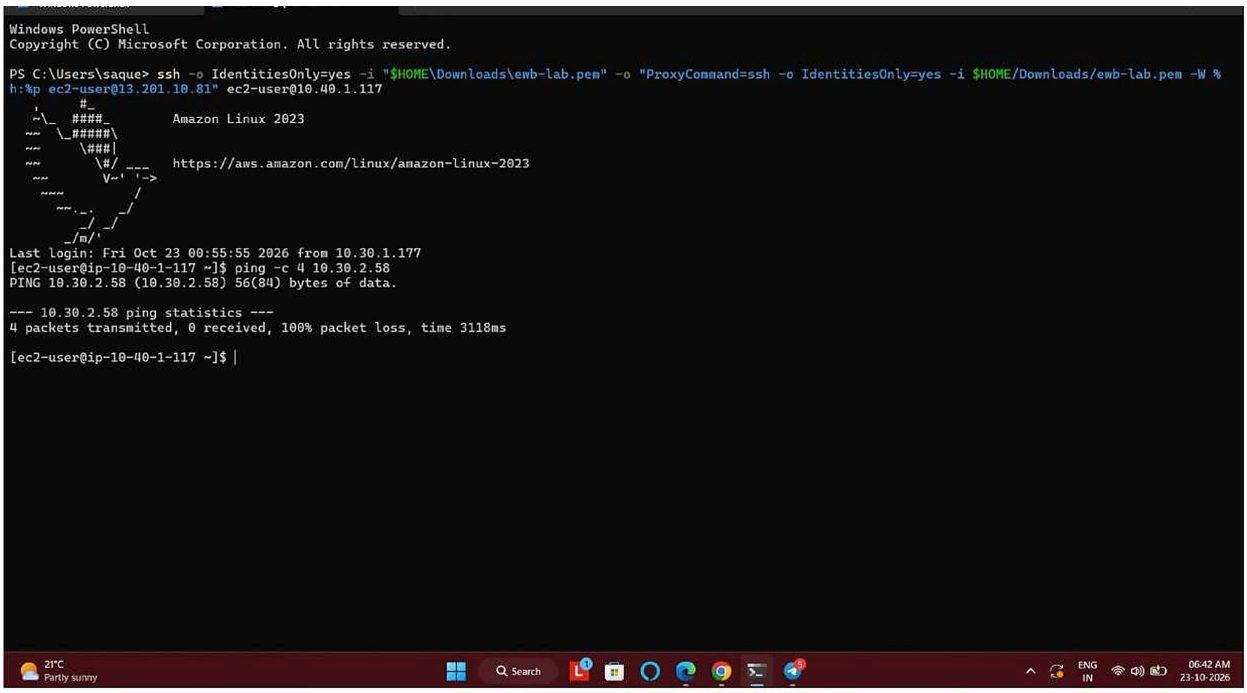

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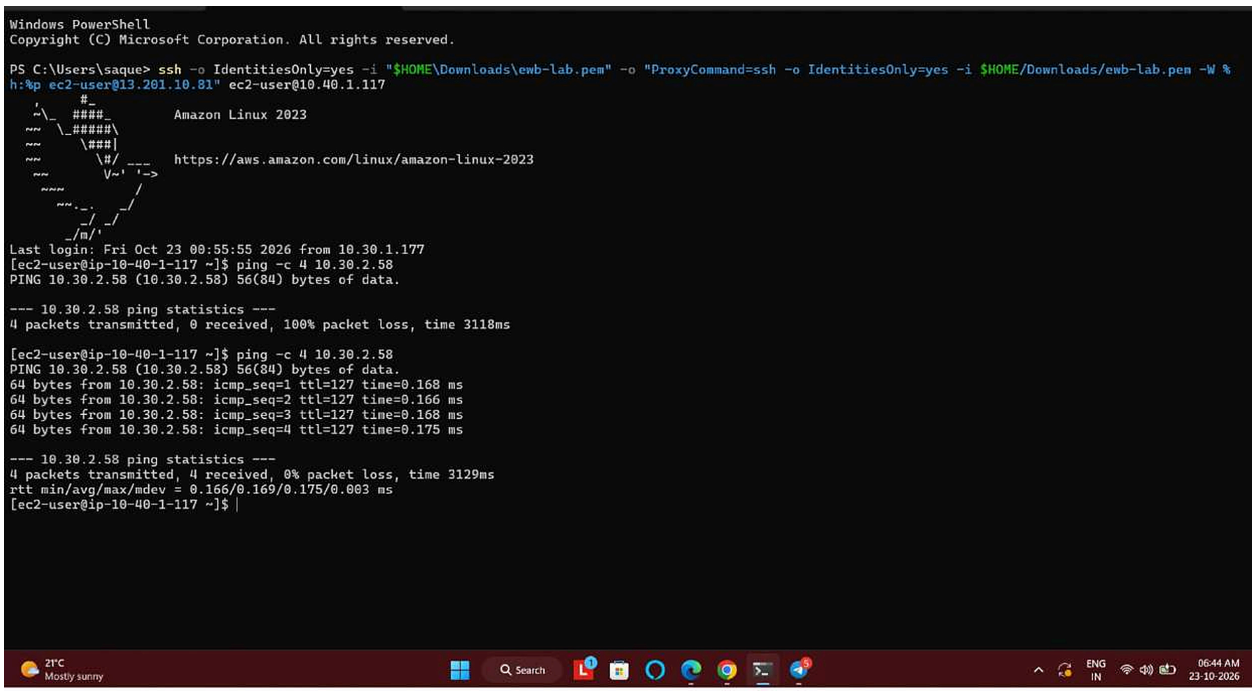

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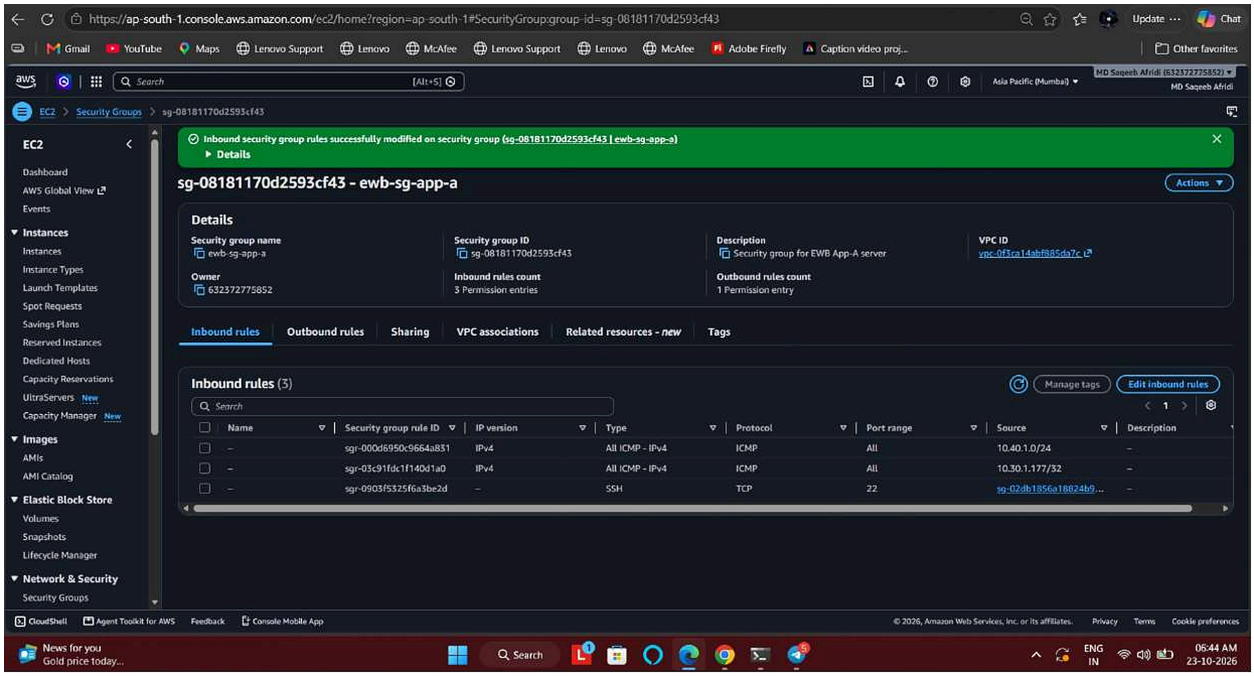

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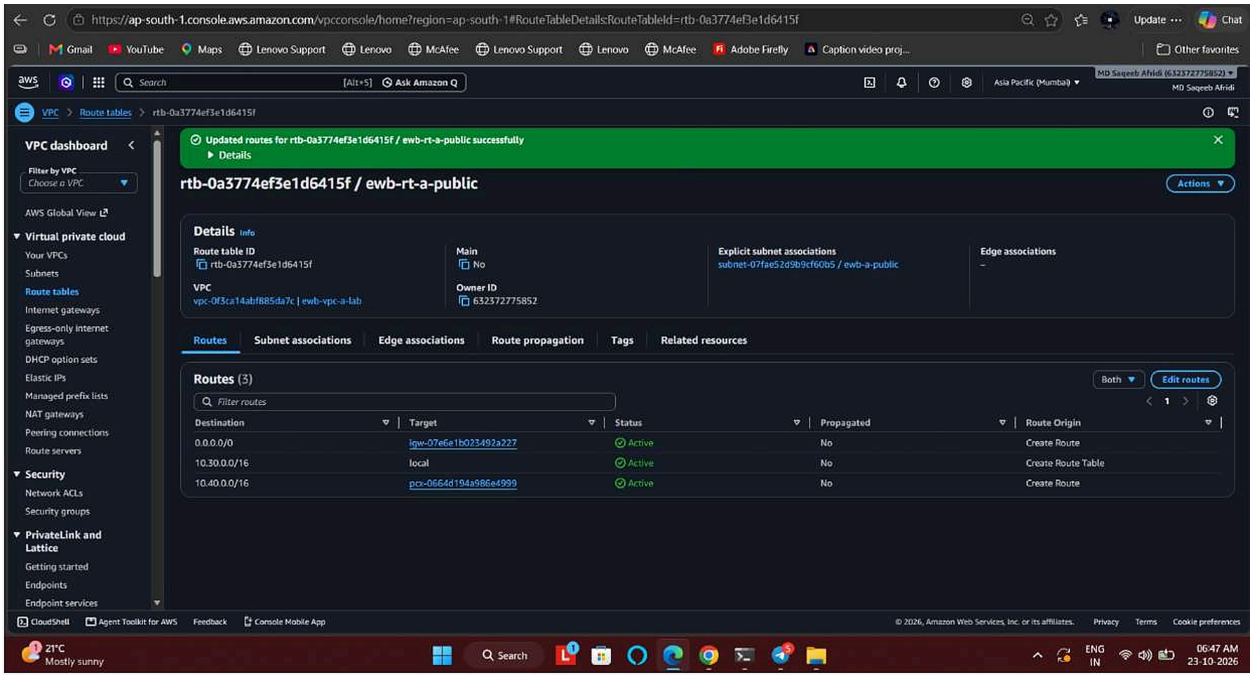

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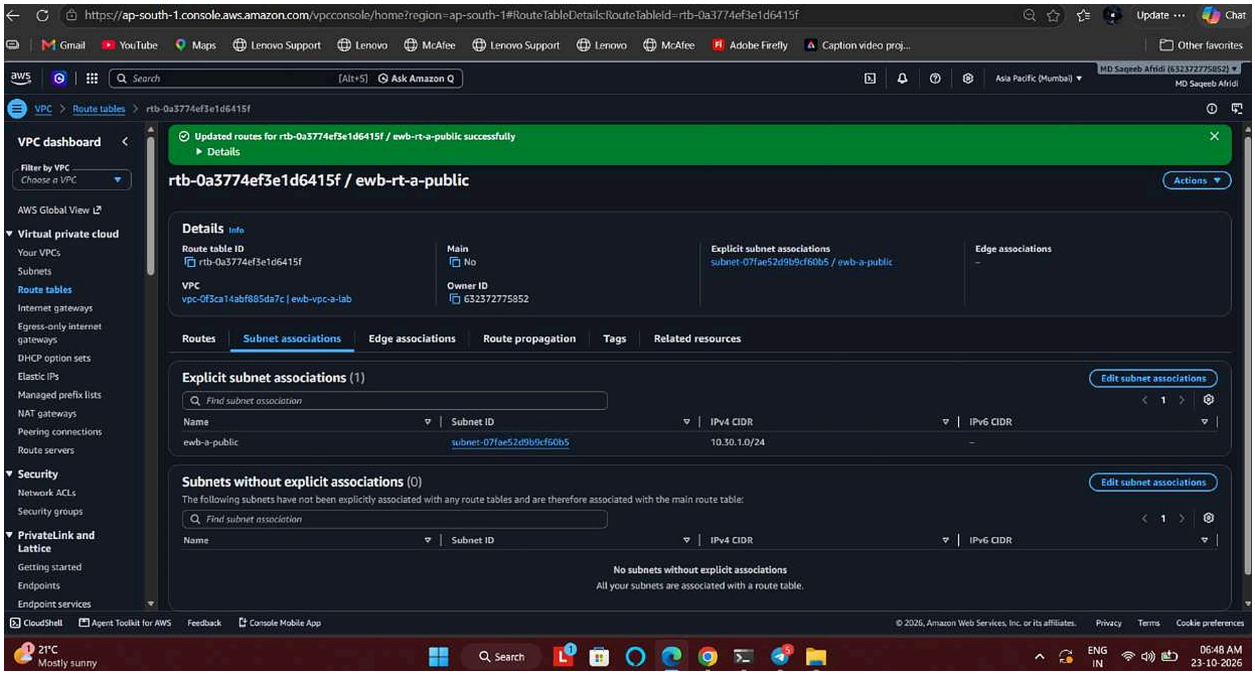

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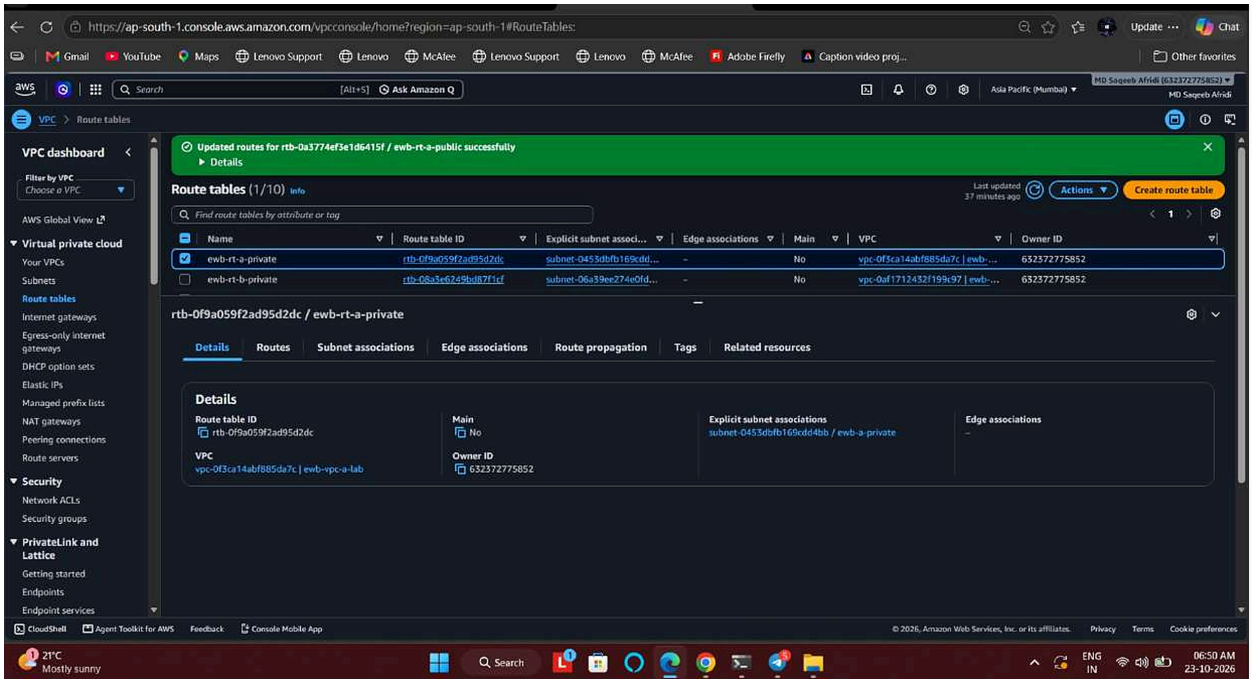

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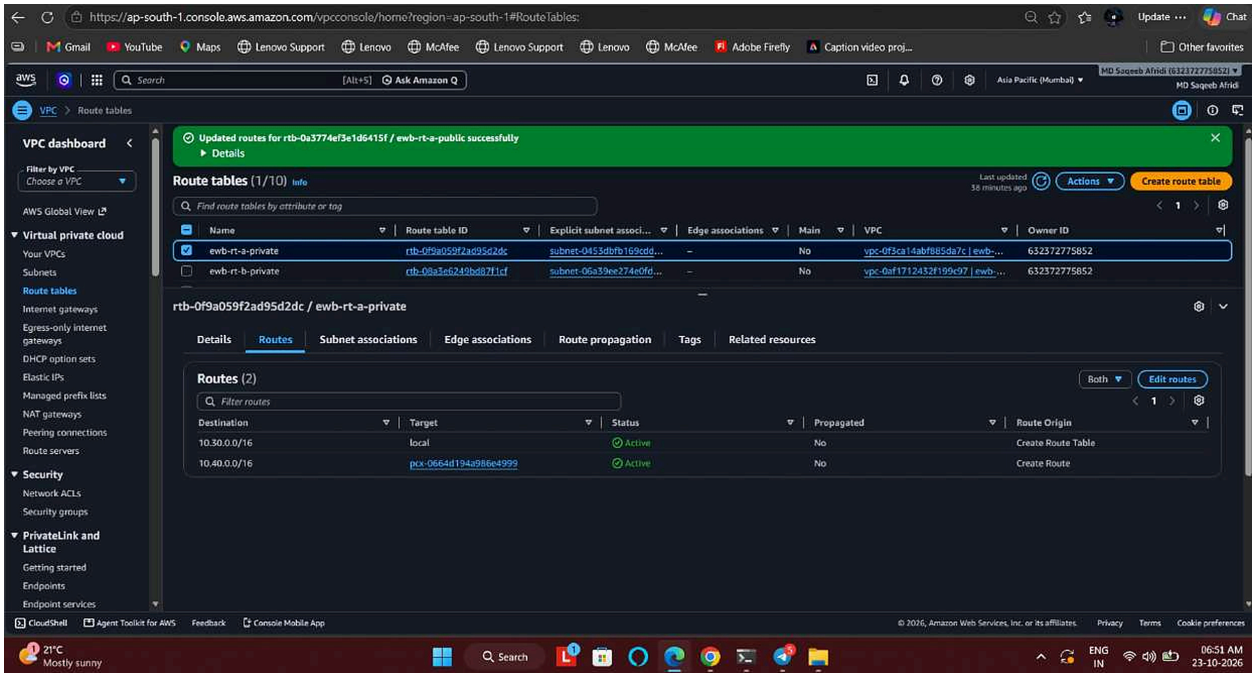

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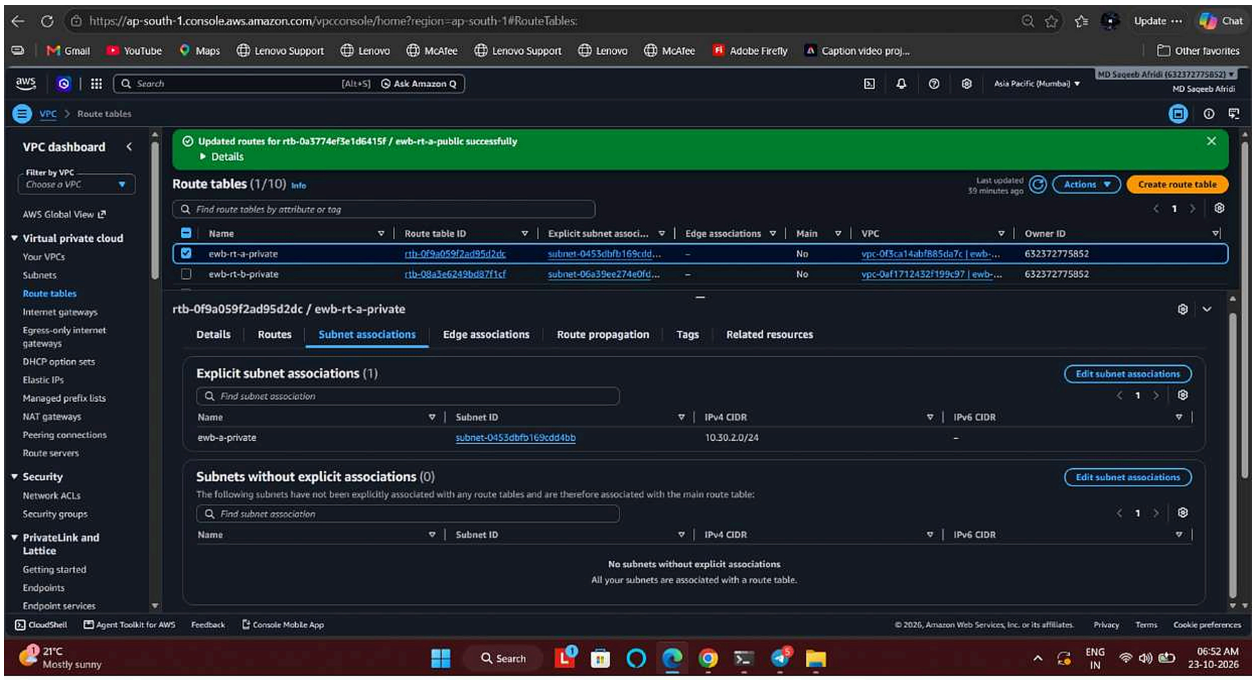

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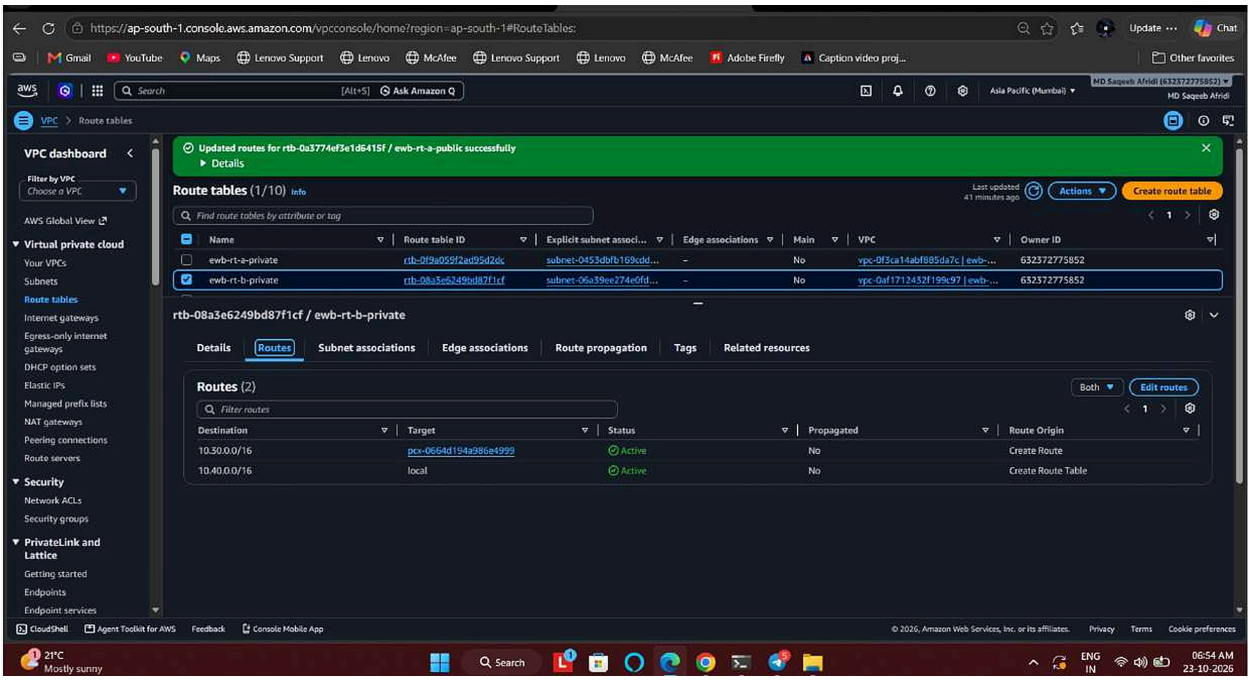

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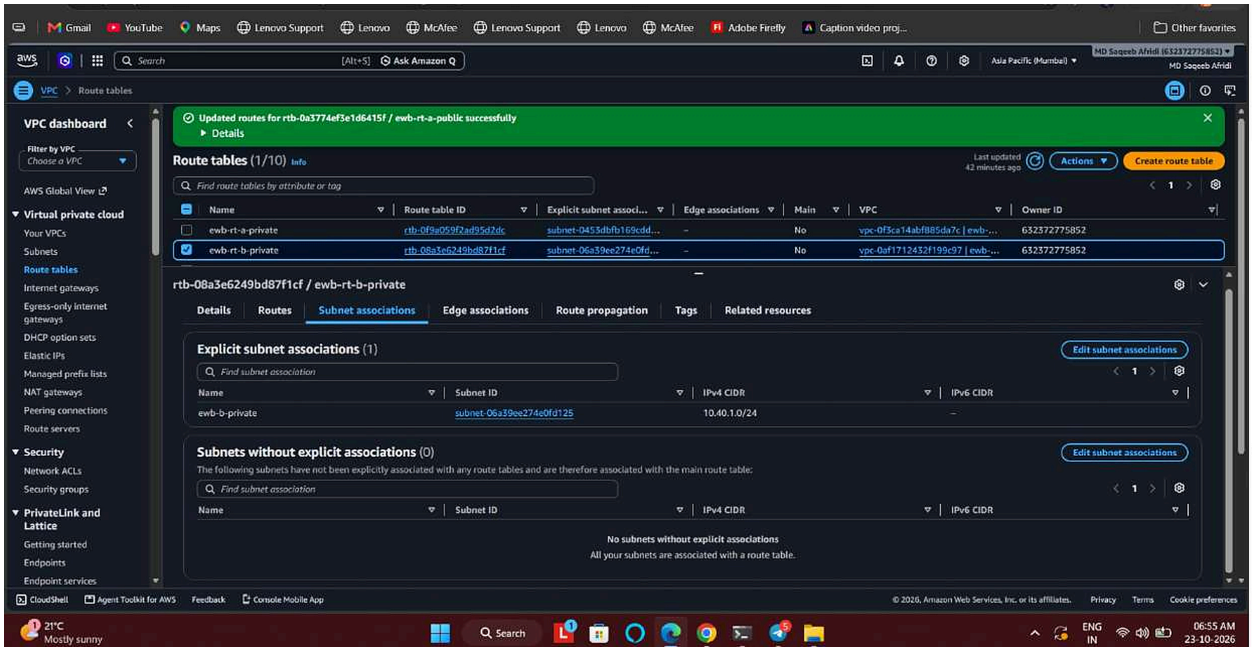

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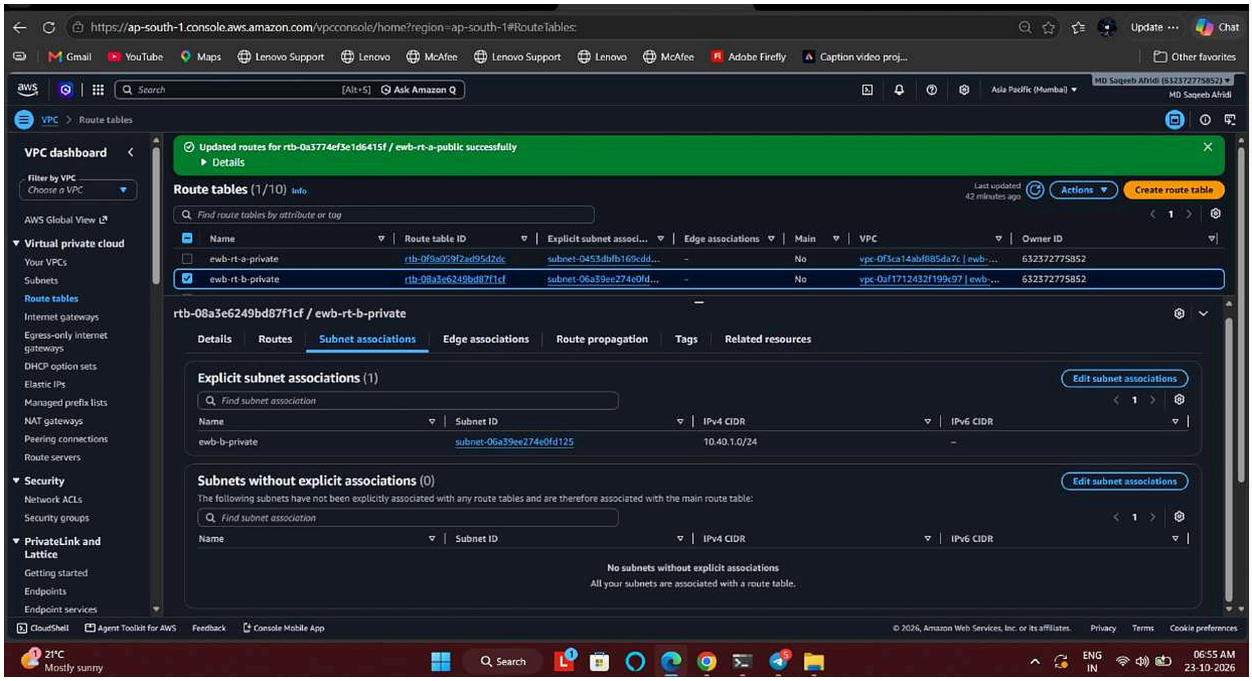

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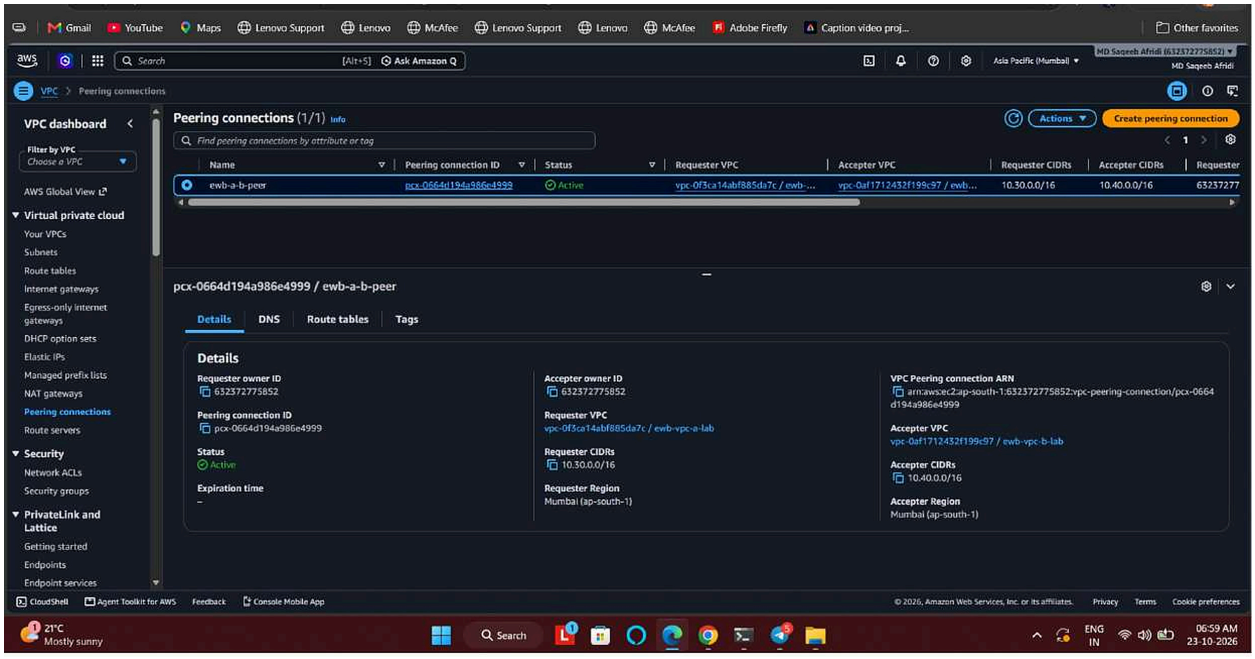

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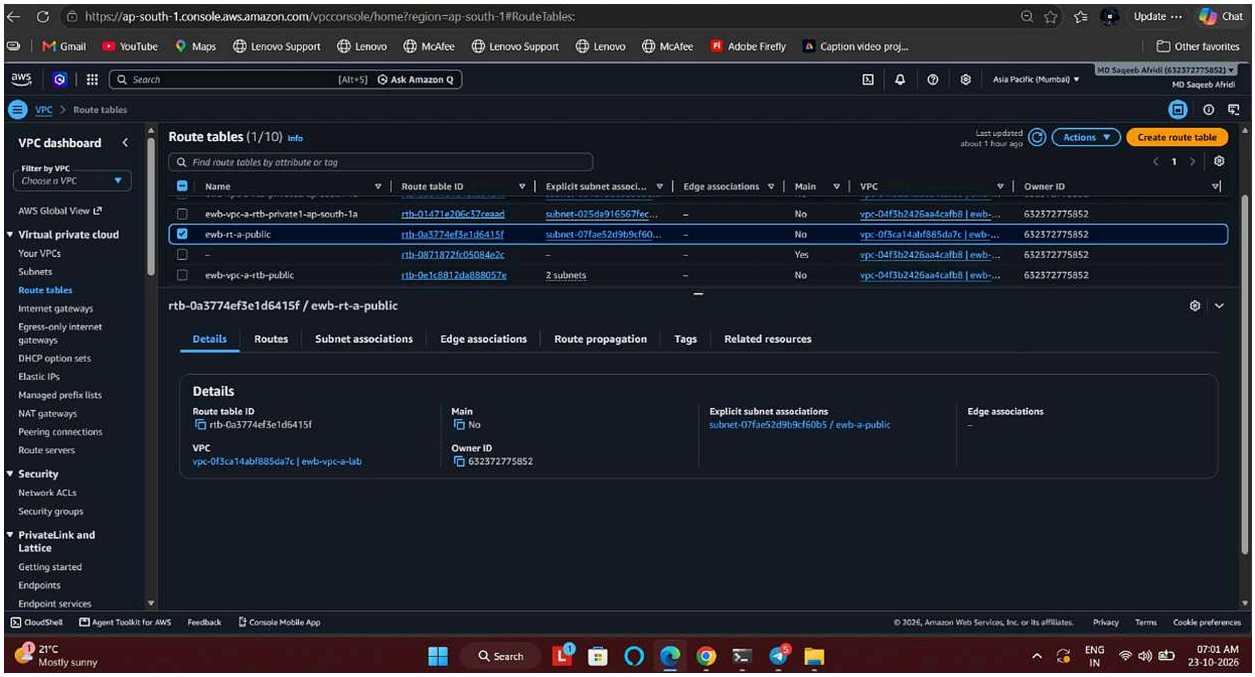

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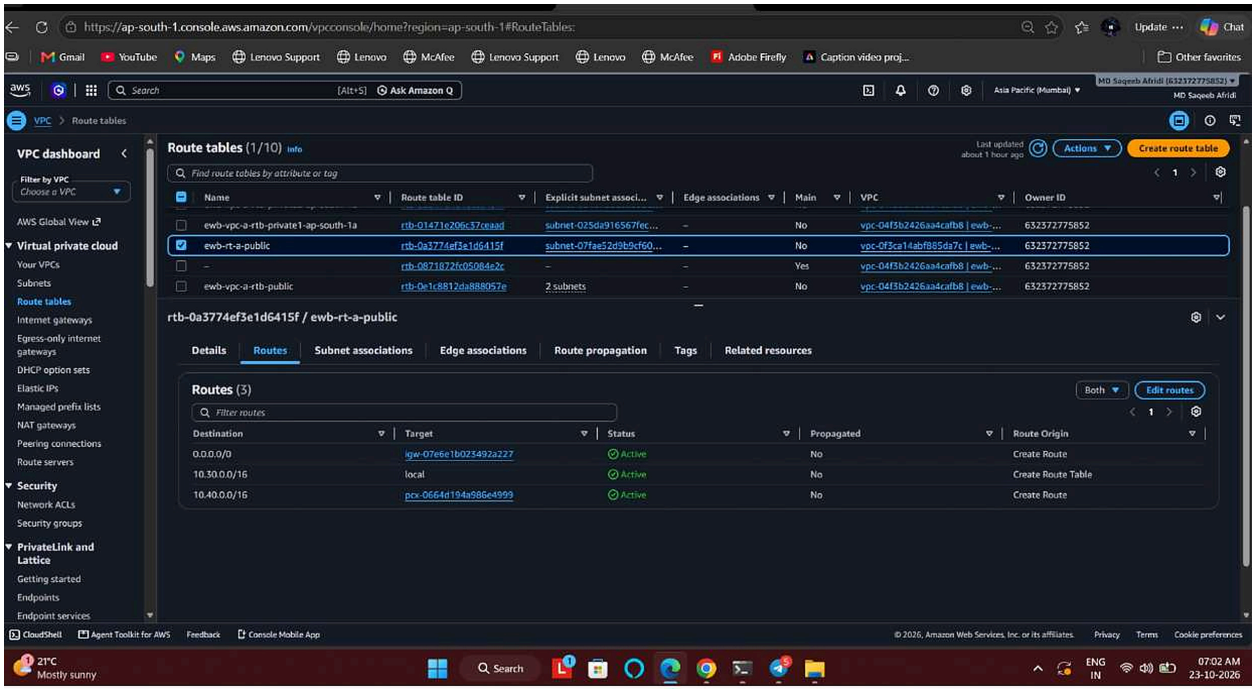

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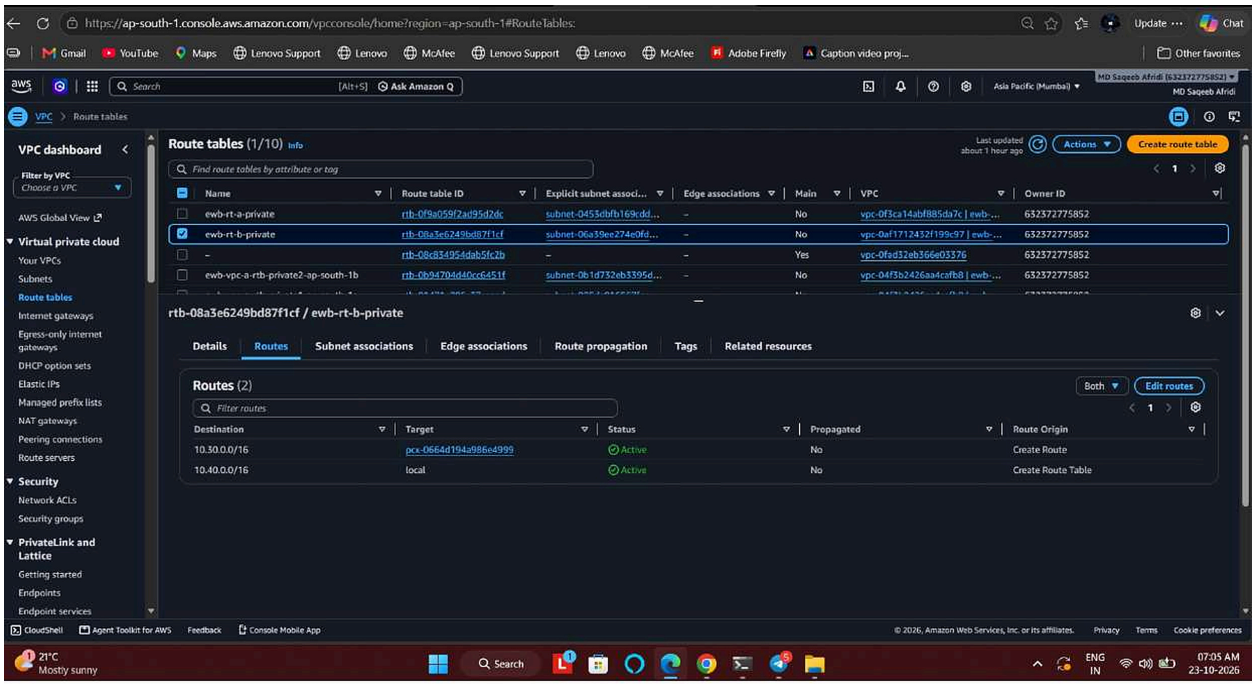
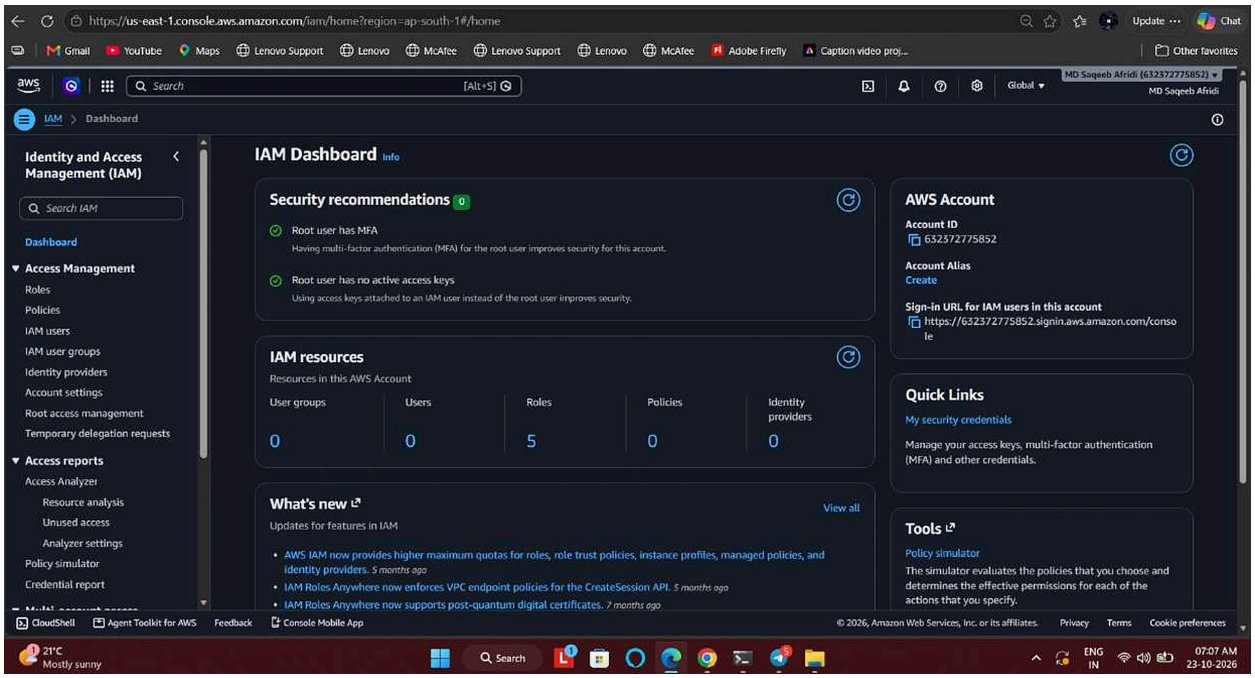
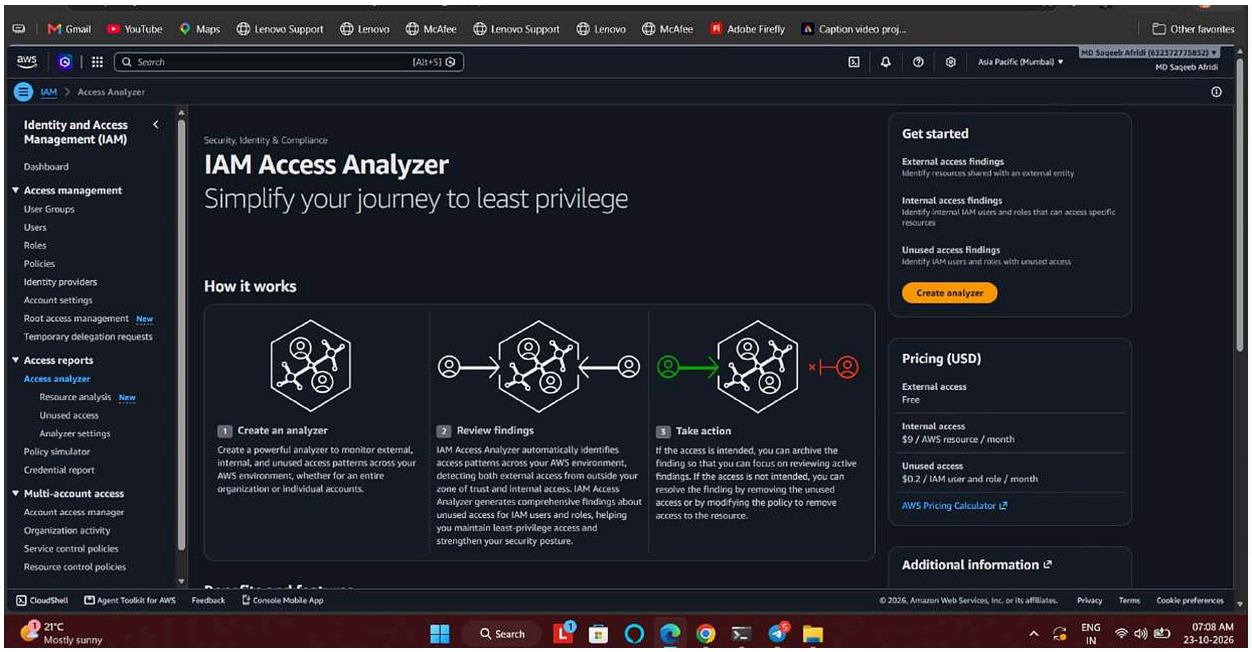


Evidence capture 40



Evidence capture 41



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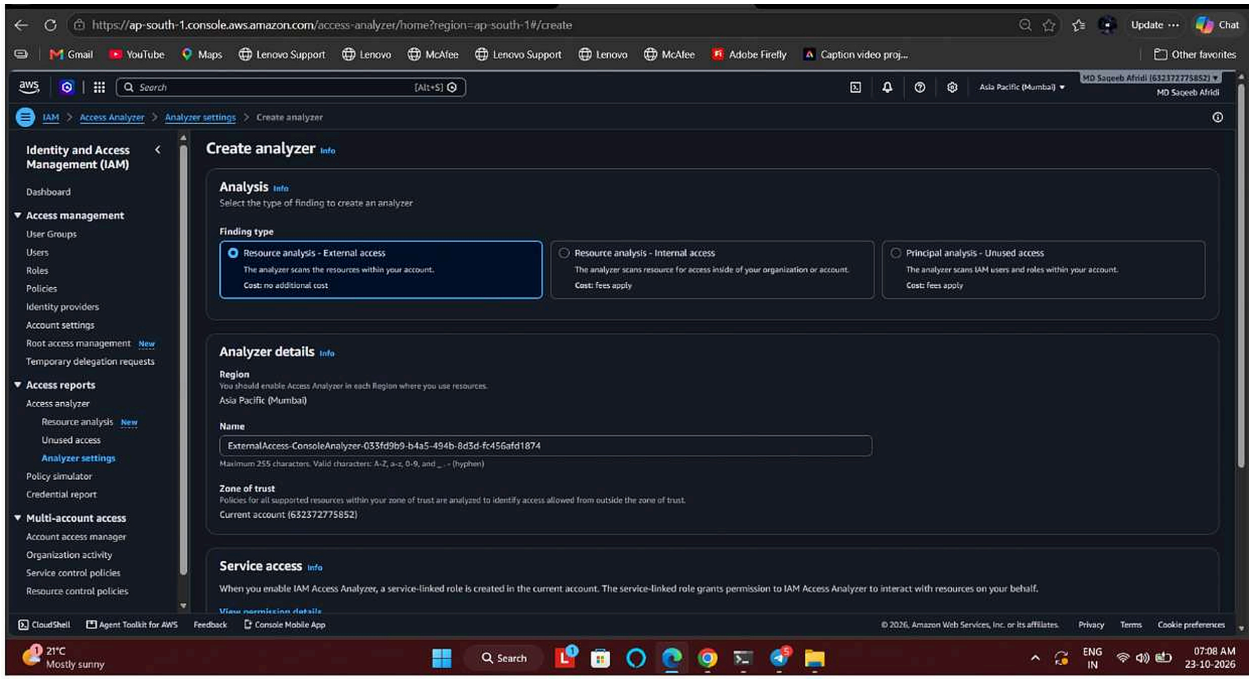
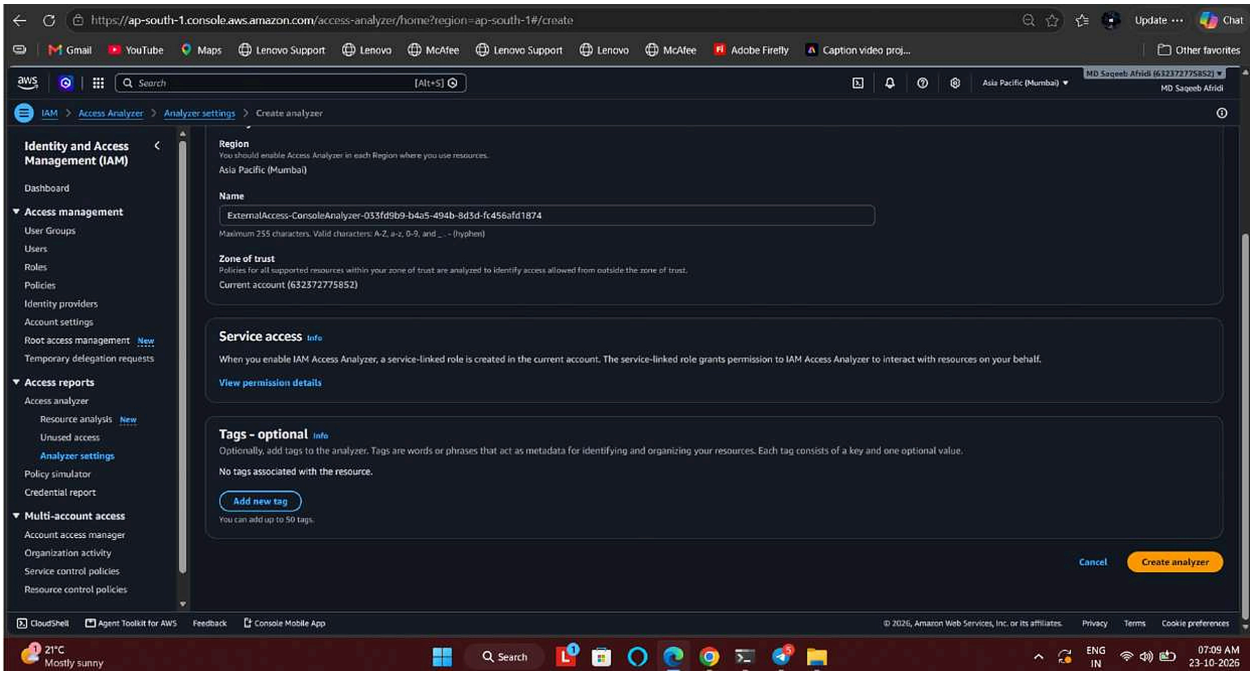
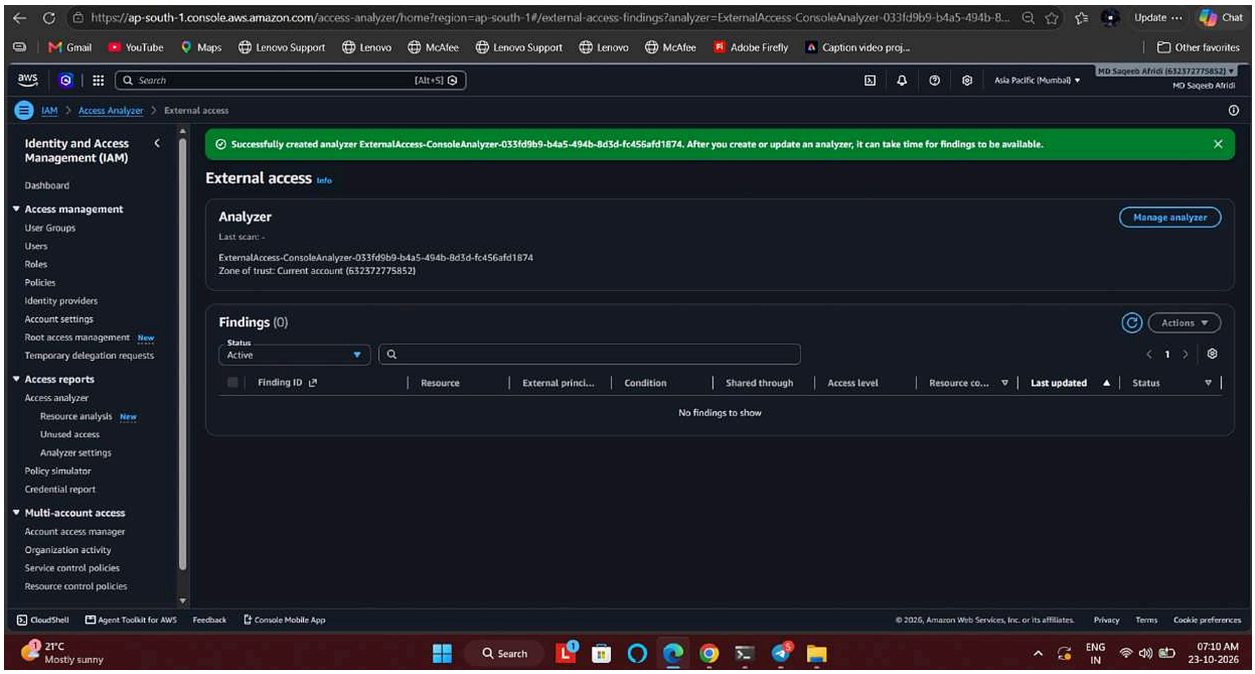


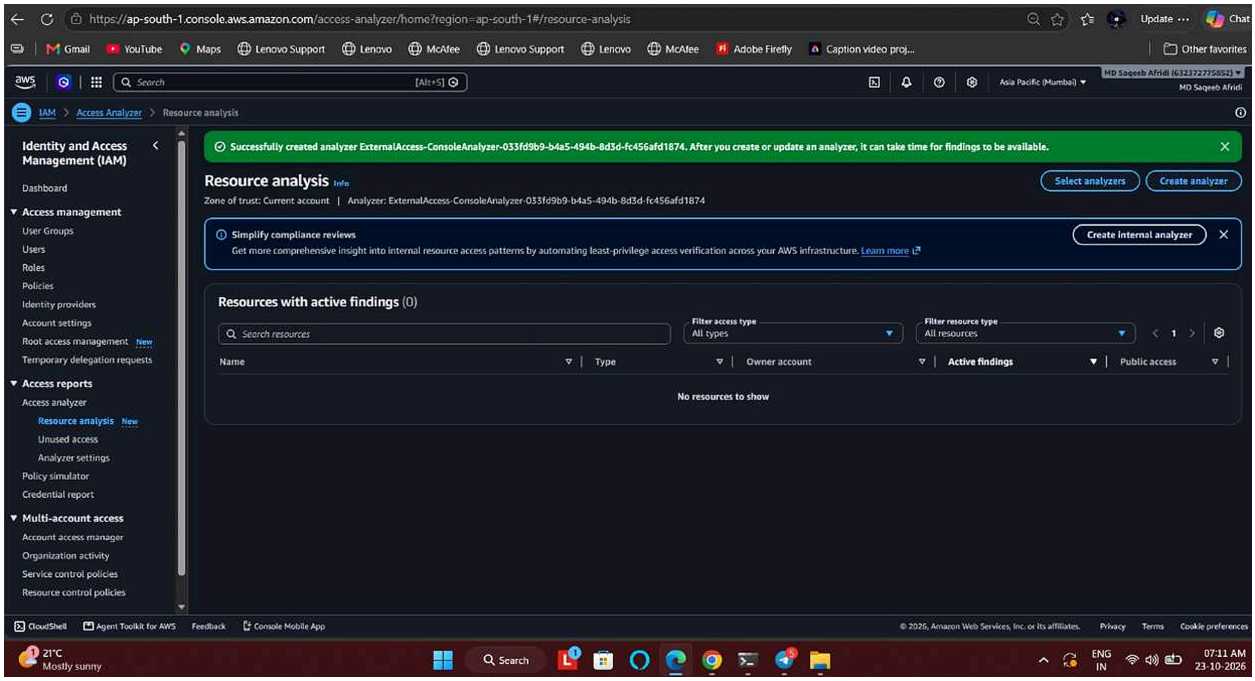
Evidence capture 43



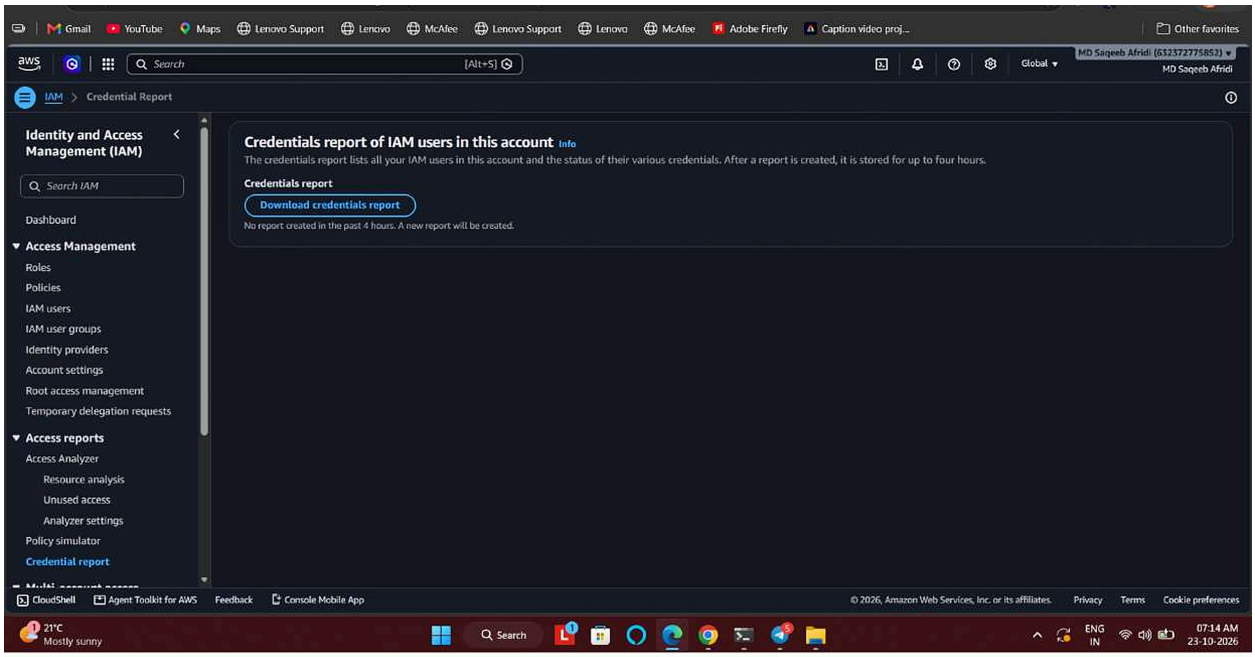
Evidence capture 44



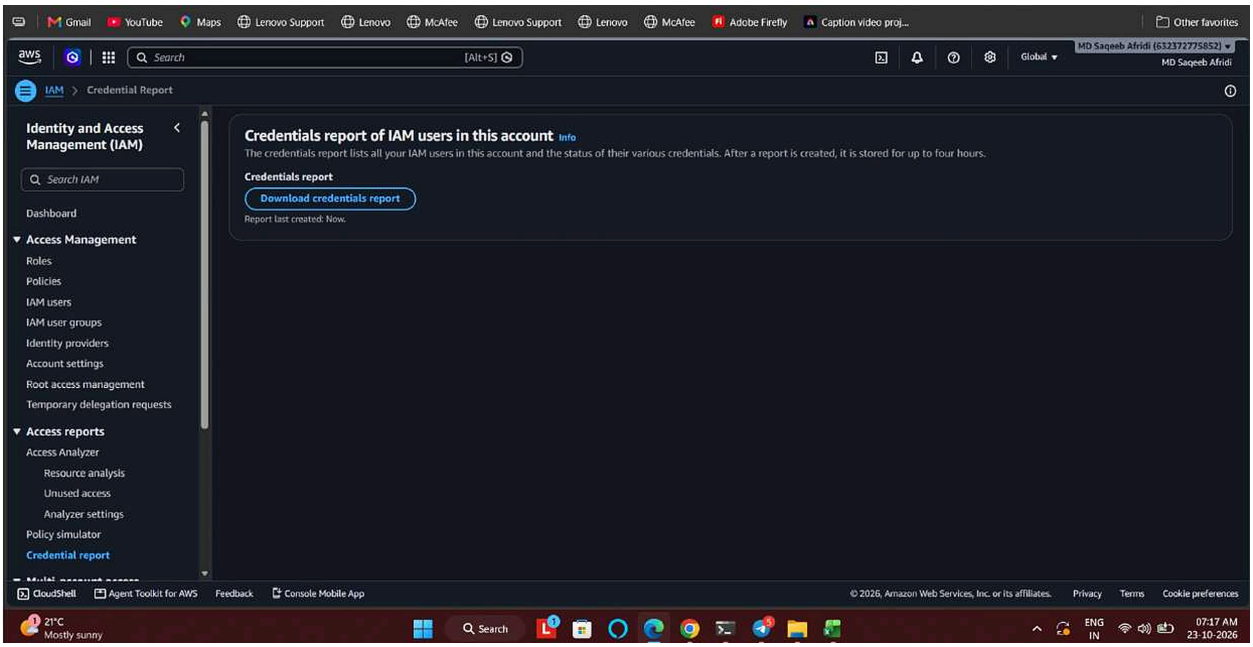
Evidence capture 45



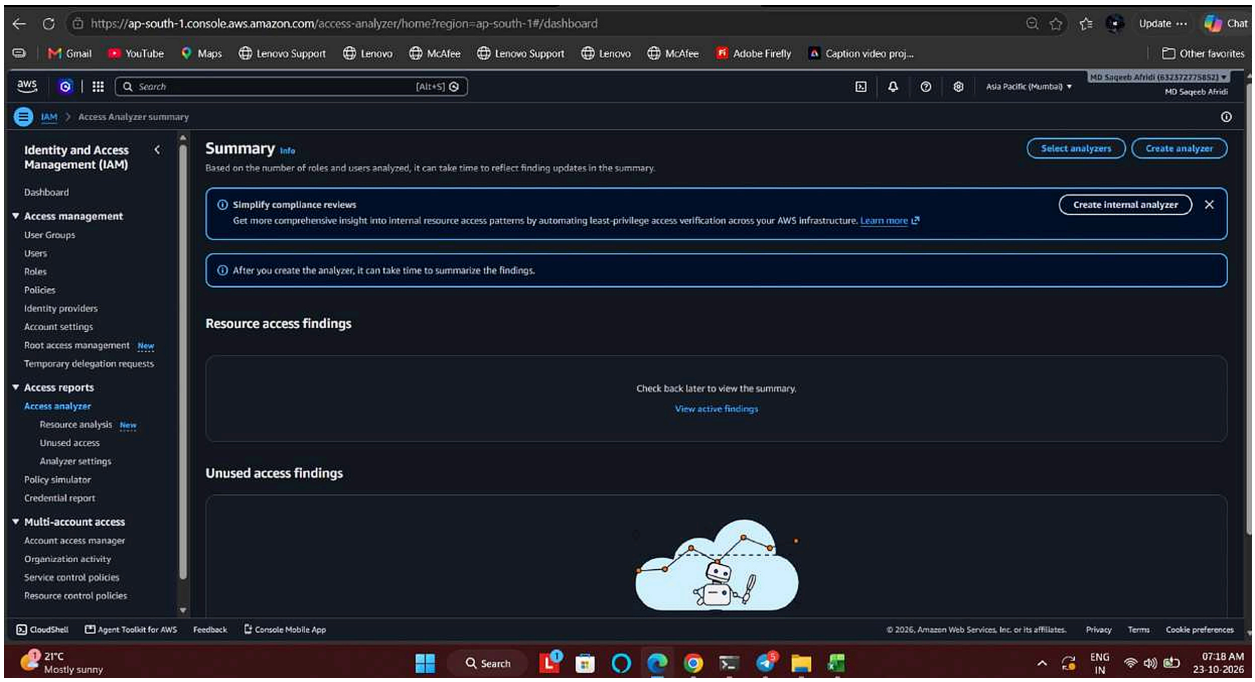
Evidence capture 46



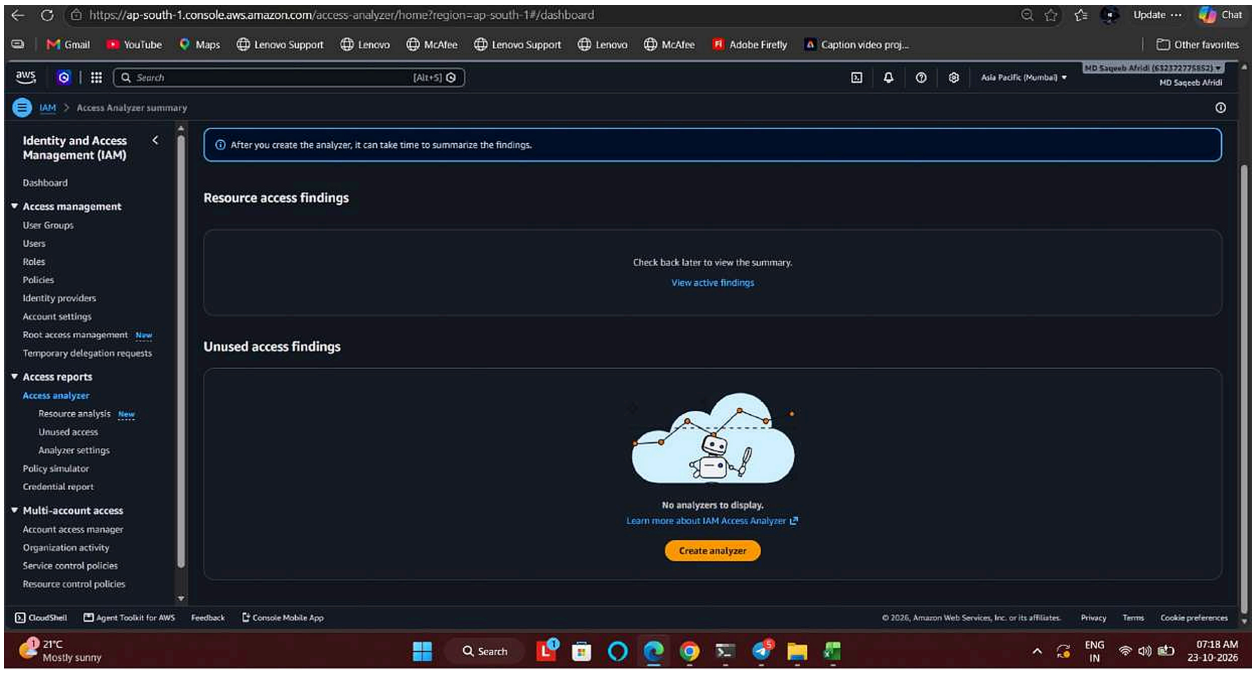
Evidence capture 47



Evidence capture 48



Evidence capture 49



Evidence capture 50